

DERIVATIVES PRICING

A COMPUTATIONAL & THEORETICAL FRAMEWORK

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Contents

I	Mathematical Foundations	6
1	Measure and Integration	6
2	Probability	6
2.1	Modelling Random Experiments	6
2.2	Random Variables	8
2.3	Probability Distributions	9
2.4	Expectation	12
2.5	L^p Spaces	14
2.6	Integral Transforms	15
2.7	Information and Independence	17
2.8	Important Stochastic Processes	19
3	Convergence	22
3.1	Motivation	22
3.2	Almost Sure Convergence	22
3.3	Convergence in Probability	23
3.4	Convergence in Distribution	24
3.5	Convergence in L^p Norm	25
3.6	Relations Between Modes of Convergence	27
3.7	Law of Large Numbers and Central Limit Theorem	27
4	Conditioning	29
4.1	Conditional Expectation	29
4.2	Conditional Probability and Distribution	31
4.3	Existence of Probability Spaces	33
4.4	Markov Property	34

5	Martingales	37
5.1	Stopping Times	37
5.2	Martingales	38
5.3	Optional Stopping	40
5.3.1	Stochastic Integration	40
5.3.2	Doob's Stopping Theorem	45
5.4	(Sub) Martingale Convergence	45
5.4.1	Upcrossing	46
5.5	Applications	48
5.5.1	Kolmogorov's 0–1 Law	49
5.5.2	Radon-Nikodym Theorem	49
6	Wiener and Brownian Motion Processes	53
6.1	Wiener Process	53
6.2	Existence	57
6.3	Strong Markov Property	58
7	Itô Calculus	60
II	Levy Processes	60
III	Stochastic Volatility	60
IV	Risk-Neutral Valuation	60
8	Discrete-time Market Model	60
9	Continuous-time Market Model	60
V	Martingale Methods	60

10 The Black-Scholes Model	60
11 A Dividend-paying Stock	60
12 Discretisation Schemes	61
13 Connection to Partial Differential Equations	61
 VI Monte Carlo Methods	 61
14 Generating Random Numbers and Random Variables	62
15 Quasi Monte Carlo	62
15.1 Discrepancy Measures	62
15.2 Low-discrepancy Sequences	62
15.2.1 Halton and Hammersley	62
15.2.2 Faure	62
15.2.3 Sobol	62
15.3 Koksma-Hlawka	62
16 Variance Reduction	62
 VII Grid & Lattice Methods	 62
17 Lattice Methods	62
18 Finite Difference Methods	63
19 Contract Sensitivities	63
19.1 Likelihood Ratio	63
19.2 Pathwise Derivatives	63
19.3 Finite-Difference Approximations	63

VIII Financial Derivatives	63
20 European Options	63
21 American Options	63
21.1 Perpetual American Put	64
21.1.1 Price under Arbitrary Exercise	66
21.1.2 Price under Optimal Exercise	70
21.1.3 Analytical Characterisation of the Put Price	73
21.1.4 Probabilistic Characterisation of the Put Price	75
21.2 Finite-Expiration American Put	80
21.2.1 Analytical Characterisation of the Put Price	80
21.2.2 Probabilistic Characterisation of the Put Price	80
21.3 American Call	80
21.3.1 Underlying Asset Pays No Dividends	81
21.3.2 Underlying Asset Pays Dividends	84
21.4 Approximate Continuation Values	90
22 Exotic Options	94
23 Basket Options	94
A Existence of Lebesgue Measure	95
B Linear Algebra and Functional Analysis	95
C Multivariate Differentiation	95
D Probability and Statistics	95
E Python Primer	95

Part I

Mathematical Foundations

1 Measure and Integration

2 Probability

2.1 Modelling Random Experiments

Probability theory is about modelling and analysing *random experiments*: experiments whose outcome cannot be determined in advance, but are nevertheless still subject to analysis. Mathematically, we can model a random experiment by defining a specific measure space $(\Omega, \mathcal{H}, \mathbb{P})$, which we call a *probability space*, where the three components are: a *sample space*, a collection of *events*, and a *probability measure*.

The *sample space* Ω of a random experiment is the set of all possible outcomes of the random experiment. Sample spaces can be countable, such as $\mathbb{N}$, or uncountable, such as $\mathbb{R}^n$ or $\{0, 1\}^\infty$. The sets in the σ -algebra $\mathcal{H}$ are called *events*. These are the subsets to which we wish to assign a probability. We can view $\mathcal{H}$ as the family of events. An event A *occurs* if the outcome of the experiment is one of the elements in A . The properties of $\mathcal{H}$ are natural in the context of random experiments and events:

- If A and B are events, the set $A \cup B$ is also an event, namely the event that A *or* B or both occur. For a sequence $A_1, A_2, \dots$ of events, their union is the event that *at least one* of the events occurs.
- If A and B are events, the set $A \cap B$ is also an event, namely the event that A *and* B both occur. For a sequence $A_1, A_2, \dots$ of events, their intersection is the event that *all* of the events occur.
- If A is an event, its complement A^c is also an event, namely the event that A *does not* occur.

- The set Ω itself is an event, namely the *certain event* (it always occurs). Similarly, $\emptyset$ is an event, namely the *impossible event* (it never occurs).

When the sample space Ω is $\mathbb{R}$, the natural σ -algebra $\mathcal{H}$ is the *Borel σ -algebra* $\mathcal{B}$. Recall that this is the smallest σ -algebra on $\mathbb{R}$ that contains all the intervals of the form $(-\infty, x]$ for $x \in \mathbb{R}$. This σ -algebra of sets is big enough to contain all important sets and small enough to allow us to assign a probability to all events.

The third ingredient in the model for a random experiment is the specification of the probability of the events. In fact, this is the crucial part of the model. Mathematically, we are looking for a *measure* $\mathbb{P}$ that assigns to each event A a number $\mathbb{P}(A)$ in $[0, 1]$ describing how likely or probable that event is.

Definition 2.1: Probability Measure

A *probability measure* is a measure on $(\Omega, \mathcal{H})$ with $\mathbb{P}(\Omega) = 1$. Thus, $\mathbb{P}$ is a mapping from $\mathcal{H}$ to $[0, 1]$ with the following properties:

- (1) $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(\Omega) = 1$.
- (2) For any sequence $A_1, A_2, \dots$ of disjoint events,

$$\mathbb{P}(\cup_n A_n) = \sum_n \mathbb{P}(A_n). \quad (2.1)$$

An event A is said to occur *almost surely* (a.s.) if $\mathbb{P}(A) = 1$.

Theorem 2.1: Properties of a Probability Measure

Let $\mathbb{P}$ be a probability measure on a measurable space $(\Omega, \mathcal{H})$, and let A, B and $A_1, A_2, \dots$ be events. Then, the following hold:

- (1) (Monotonicity): $A \subseteq B \implies \mathbb{P}(A) \leq \mathbb{P}(B)$.
- (2) (Continuity from below): If $A_1, A_2, \dots$ is an increasing sequence of events, i.e., $A_1 \subseteq A_2 \subseteq \dots$, then

$$\lim \mathbb{P}(A_n) = \mathbb{P}(\cup_n A_n). \quad (2.2)$$

- (3) (Countable subadditivity): $\mathbb{P}(\cup_n A_n) \leq \sum_n \mathbb{P}(A_n)$.

Theorem 2.2: Additional Properties of a Probability Measure

Let $\mathbb{P}$ be a probability measure on a measurable space $(\Omega, \mathcal{H})$, and let A, B and $A_1, A_2, \dots$ be events. Then, the following hold:

- (1) (Complement): $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
- (2) (Union): $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
- (3) (Continuity from above): If $A_1, A_2, \dots$ is a decreasing sequence of events, i.e., $A_1 \supseteq A_2 \supseteq \dots$, then

$$\lim \mathbb{P}(A_n) = \mathbb{P}(\cap_n A_n). \quad (2.3)$$

2.2 Random Variables

Usually, the most convenient way to describe quantities of interest connected with random experiments is by using random variables. Random variables allow us to use intuitive notations for certain events, such as $\{X \in A\}$, etc. Intuitively, we can think of a random variable X as a measurement on a random experiment that will become available *tomorrow*. However, all the think work can be done *today*. If for each possible outcome ω , the measurement is $X(\omega)$, we can specify today probabilities such as $\mathbb{P}(X \in A)$.

We can translate our intuitive notion of a random variable into rigorous mathematics by defining a random variable to be a measurable function from $(\Omega, \mathcal{H})$ to a measurable space $(E, \mathcal{E})$.

Definition 2.2: Random Variable

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and $(E, \mathcal{E})$ a measurable space. A *random variable* X taking values in E is an $\mathcal{H}/\mathcal{E}$ -measurable function; that is, a mapping $X : \Omega \rightarrow E$ that satisfies

$$X^{-1}A = \{X \in A\} = \{\omega \in \Omega : X(\omega) \in A\} \in \mathcal{H} \quad \text{for all } A \in \mathcal{E}. \quad (2.4)$$

Often a random experiment is described via more than one random variable. Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and let $\mathbb{T}$ be an arbitrary index set, which may be countable or uncountable. Think of $\mathbb{T}$ as a set of times. For each $t \in \mathbb{T}$, let $(E_t, \mathcal{E}_t)$ be a measurable

space. The product measurable space of $\{(E_t, \mathcal{E}_t) : t \in \mathbb{T}\}$ is

$$\left(\times_{t \in \mathbb{T}} E_t, \bigotimes_{t \in \mathbb{T}} \mathcal{E}_t \right).$$

Each element $x \in \times_{t \in \mathbb{T}} E_t$ is of the form $x = (x_t)_{t \in \mathbb{T}}$. Often $(E_t, \mathcal{E}_t)$ is the same space $(E, \mathcal{E})$ for all t , in which case x can be viewed as a function $x : \mathbb{T} \rightarrow E$. The σ -algebra on $\times_{t \in \mathbb{T}} E_t$ is generated by the rectangles

$$R((A_t)_{t \in \mathbb{T}}) := \left\{ x \in \times_{t \in \mathbb{T}} E_t : x_t \in A_t \text{ for each } t \in \mathbb{T} \right\}, \quad (2.5)$$

where $A_t \in \mathcal{E}_t$ and $A_t = E_t$ except for finitely many t .

Definition 2.3: Stochastic Process

A collection of random variables $X := \{X_t, t \in \mathbb{T}\}$, where $\mathbb{T}$ is any index set, is called a *stochastic process*. When $\mathbb{T}$ is finite, X is called a *random vector*.

Theorem 2.3: Stochastic Process

$X := \{X_t, t \in \mathbb{T}\}$ is a stochastic process taking values in $\times_{t \in \mathbb{T}} E_t$ with σ -algebra $\bigotimes_{t \in \mathbb{T}} \mathcal{E}_t$ if and only if each X_t is a random variable taking values in E_t with σ -algebra $\mathcal{E}_t$.

2.3 Probability Distributions

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and let X be a random variable taking values in a set E with σ -algebra $\mathcal{E}$. To simplify our usage, we also say that “ X takes values in $(E, \mathcal{E})$ ”. If we wish to describe our experiment via X , then we need to specify the probabilities of events such as $\{X \in B\}$, $B \in \mathcal{E}$. In elementary probability theory we usually made a distinction between “discrete” and “continuous” random variables. However, it is sometimes better to analyse random variables in a more unified manner; many definitions and properties of random variables do not depend on whether they are “discrete” or “continuous”. Moreover, some random variables are neither discrete nor continuous.

Recall that by definition *all* the sets $\{X \in B\}$ with $B \in \mathcal{E}$ are events. So we may assign a probability to each of these events. This leads to the notion of the *distribution* of a random variable.

Definition 2.4: Probability Distribution

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and let X be a random variable taking values in $(E, \mathcal{E})$. The (*probability*) *distribution* of X is the image measure μ of $\mathbb{P}$ under X . That is,

$$\mu(B) := \mathbb{P}(X \in B), \quad B \in \mathcal{E}. \quad (2.6)$$

Definition 2.5: Cumulative Distribution Function

The (*cumulative*) *distribution function* of a numerical random variable X is the function $F : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F(x) := \mathbb{P}(X \leq x), \quad x \in \mathbb{R}. \quad (2.7)$$

Proposition 2.1: Properties of a Cumulative Distribution Function

- (1) (*Bounded*): $0 \leq F(x) \leq 1$.
- (2) (*Increasing*): $x \leq y \implies F(x) \leq F(y)$.
- (3) (*Right-continuous*): $\lim_{h \downarrow 0} F(x + h) = F(x)$.

Definition 2.6: Discrete Distribution

We say that X has a *discrete distribution* μ on $(E, \mathcal{E})$ if μ is a discrete measure; that is, it is of the form

$$\mu = \sum_{x \in D} f(x) \delta_x$$

for some countable set D and positive masses $\{f(x), x \in D\}$.

Definition 2.7: Distribution with a Density

We say that X has an *absolutely continuous* distribution μ on $(E, \mathcal{E})$ with respect to a measure λ on $(E, \mathcal{E})$ if there exists a positive $\mathcal{E}$ -measurable function f such that $\mu = f\lambda$; that is,

$$\mu(B) = \mathbb{P}(X \in B) = \int_B \lambda(dx) f(x), \quad B \in \mathcal{E}.$$

The function f is thus the (*probability*) *density function* (pdf) of X with respect to the measure, λ often taken to be the Lebesgue measure. Note that μ is absolutely continuous with respect to λ in the sense that for all $B \in \mathcal{E} : \lambda(B) = 0 \implies \mu(B) = 0$.

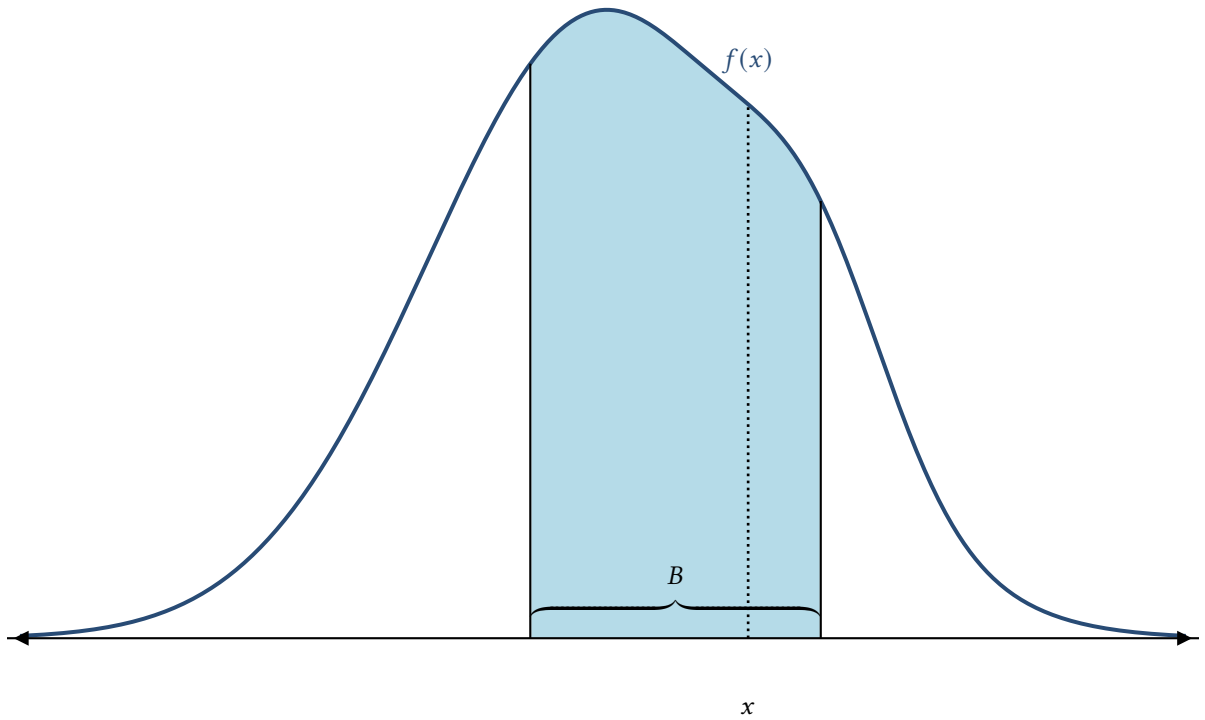


Figure 1: A probability density function of a numerical random variable

Describing an experiment via a random variable and its probability distribution or density is often more convenient than specifying the probability space. In fact, the probability space usually stays in the background. The question remains, however, whether there *exists* a probability space $\Omega, \mathcal{H}, \mathbb{P}$ and a numerical random variable X for a given probability distribution. Fortunately, existence can be established for all practical probability models.

The concept of independence is of great importance in the study of random experiments and the construction of probability distributions. Loosely speaking, independence is about the lack of shared information between random objects.

Definition 2.8: Independent Random Variables

Let X and Y be random variables taking values in $(E, \mathcal{E})$ and $(F, \mathcal{F})$, respectively. They are said to be *independent* if for any $A \in \mathcal{E}$ and $B \in \mathcal{F}$ it holds that

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B). \quad (2.8)$$

Intuitively, this means that information regarding X does not affect our knowledge of Y and vice versa. Mathematically, Equation (2.8) simply states that the distribution, μ say, of (X, Y) is given by the *product measure* of the distributions μ_X of X and μ_Y of Y . We can extend the independence concept to a finite or infinite collection of random variables as follows:

Definition 2.9: Independency

We say that a collection of random variables $\{X_t, t \in \mathbb{T}\}$, with each X_t taking values in $(E_t, \mathcal{E}_t)$, is an *independency* if for any finite choice of indexes $t_1, \dots, t_n \in \mathbb{T}$ and sets $A_{t_1} \in \mathcal{E}_{t_1}, \dots, A_{t_n} \in \mathcal{E}_{t_n}$ it holds that

$$\mathbb{P}(X_{t_1} \in A_{t_1}, \dots, X_{t_n} \in A_{t_n}) = \mathbb{P}(X_{t_1} \in A_{t_1}) \cdots \mathbb{P}(X_{t_n} \in A_{t_n}).$$

Attention 2.1: Independence as a Modelling Assumption

More often than not, the independence of random variables is a model *assumption*, rather than a consequence. Instead of describing a random experiment via an explicit description of Ω and $\mathbb{P}$, we will usually model the experiment through one or more (independent) random variables

2.4 Expectation

The expectation of a (numerical) random variable is a weighted average of all the values that the random variable can take. The expectation of a *discrete* random variable X is defined

as:

$$\mathbb{E}X := \sum_x x\mathbb{P}(X = x),$$

and for a *continuous* random variable X with pdf f it is defined as:

$$\mathbb{E}X := \int_{-\infty}^{\infty} xf(x)dx.$$

Having to define the expectation in two ways is unsatisfactory. Also, there are cases wherein random variables are neither discrete nor continuous. How should we define the expectation for these random variables? Here is the formal definition:w4

Definition 2.10: Expectation

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and X a numerical random variable. The *expectation* of X is defined as the integral

$$\mathbb{E}X := \mathbb{P}X = \int Xd\mathbb{P}.$$

Theorem 2.4: Properties of an Expectation

Let $a, b \in \mathbb{R}$ and let X, Y , and (X_n) be numerical random variables for which the expectations below are well-defined. Then, the following hold:

- (1) (Monotonicity): If $X \leq Y$, then $\mathbb{E}X \leq \mathbb{E}Y$.
- (2) (Linearity): $\mathbb{E}(aX + bY) = a\mathbb{E}X + b\mathbb{E}Y$.
- (3) (Monotone convergence): If $X_n \uparrow X$, then $\mathbb{E}X_n \uparrow \mathbb{E}X$.

Lemma 2.1: Fatou

For any sequence (X_n) of positive numerical random variables,

$$\mathbb{E} \liminf X_n \leq \liminf \mathbb{E}X_n.$$

Theorem 2.5: Dominated and Bounded Convergence

Let (X_n) be a sequence of numerical random variables for which $\lim X_n = X$ exists.

Then, the following hold:

- (1) (Dominated convergence): If, for each n , $|X_n| \leq Y$ for some random variable Y with $\mathbb{E}Y < \infty$, then $\mathbb{E}X_n \rightarrow \mathbb{E}X < \infty$.
- (2) (Bounded convergence): If $|X_n| \leq c$ for all n for some $c \in \mathbb{R}$, then $\mathbb{E}X_n \rightarrow \mathbb{E}X < \infty$.

Theorem 2.6: Expectation and Image Measure

Let X be a random variable taking values in $(E, \mathcal{E})$ with distribution μ and let h be an $\mathcal{E}$ -measurable function. Then, provided the integral exists:

$$\mathbb{E}h(X) = \int h(X) d\mathbb{P} = \int_{\mathbb{R}} \mu(dx) h(x) = \mu h.$$

Theorem 2.7: Transformation Rule

Let X be a random vector with density f_X with respect to the Lebesgue measure on $(\mathbb{R}^n, \mathcal{B}^n)$. Let $Z := g(X)$, where $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible with inverse g^{-1} and Jacobian $|\frac{\partial z}{\partial x}|$. Then, at $z = g(x)$ the random vector Z has the following density with respect to the Lebesgue measure on $(\mathbb{R}^n, \mathcal{B}^n)$:

$$f_Z(z) := \frac{f_X(x)}{|\frac{\partial z}{\partial x}|} = f_X(g^{-1}(z)) \left| \frac{\partial x}{\partial z} \right|, \quad z \in \mathbb{R}^n. \quad (2.9)$$

Proposition 2.2: Expectation of a Quadratic Form

Let A be an $n \times n$ matrix and X an n -dimensional random vector with expectation vector μ and covariance matrix Σ . The random variable $Y := X^\top A X$ has expectation $\text{tr}(\Sigma A) + \mu^\top A \mu$.

2.5 L^p Spaces

Let X be a numerical random variable on

Definition 2.11: L^p Space

For each $p \in [1, \infty]$ the space L^p is comprised of all numerical random variables X for which $\|X\|_p < \infty$.

Lemma 2.2: Jensen's Inequality

Let $h : \mathcal{X} \rightarrow \mathbb{R}$ be a convex function and let X be a random variable taking values in $\mathcal{X}$, with expectation vector $\mathbb{E}X$. Then,

$$\mathbb{E}h(X) \geq h(\mathbb{E}X).$$

Theorem 2.8: Properties of the L^p Norm

Let X and Y be numerical random variables on $(\Omega, \mathcal{H}, \mathbb{P})$. Then, the following hold:

- (1) (Positivity): $\|X\|_p \geq 0$, and $\|X\|_p = 0 \iff X = 0$ almost surely.
- (2) (Multiplication by a constant): $\|cX\|_p = |c|\|X\|_p$.
- (3) (Minkowski's (triangle) inequality):

$$|\|X\|_p - \|Y\|_p| \leq \|X + Y\|_p \leq \|X\|_p + \|Y\|_p.$$

- (4) (Hölder's inequality): For $p, q, r \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = \frac{1}{r}$,

$$\|XY\|_r \leq \|X\|_p \|Y\|_q. \quad (2.10)$$

- (5) (Monotonicity): If $1 \leq p < q \leq \infty$, then $\|X\|_p \leq \|X\|_q$.

2.6 Integral Transforms

Expectations are used in many probabilistic analyses. In this section, we highlight their use in integral transforms. Many calculations and manipulations involving the use of such transforms facilitate the analysis of probability distributions. We describe two major types of transforms.

Definition 2.12: Moment Generating Function

The *moment generating function (MGF)* of a numerical random variable X with distribution μ is the function $M : \mathbb{R} \rightarrow [0, \infty]$, given by

$$M(s) := \mathbb{E}e^{sX} = \int_{-\infty}^{\infty} \mu(dx)e^{sx}, \quad s \in \mathbb{R}.$$

Theorem 2.9: Uniqueness of the MGF

A measure μ is determined by its MGF M if the latter is finite in a neighborhood of 0.

Theorem 2.10: Taylor's Theorem for MGFs

Suppose $M(s) := \mathbb{E}e^{sX}$ is finite in the neighborhood $(-s_0, s_0)$ of 0. Then, M has the Taylor expansion

$$M(s) = \sum_{k=0}^{\infty} \frac{s^k}{k!} \mathbb{E}X^k, \quad |s| < s_0.$$

In particular, all moments $\mathbb{E}X^k, k \in \mathbb{N}$ are finite.

Theorem 2.11: Product of MGFs

If X and Y are independent, then $M_{X+Y}(s) = M_X(s)M_Y(s)$.

Definition 2.13: Characteristic Function

The *characteristic function* of a numerical random variable X with distribution μ is the function $\psi : \mathbb{R} \rightarrow \mathbb{C}$, defined by

$$\psi(r) := \mathbb{E}e^{irX} = \mathbb{E} \cos(rX) + i\mathbb{E} \sin(rX) = \int_{-\infty}^{\infty} \mu(dx)e^{irx}, \quad r \in \mathbb{R}.$$

Theorem 2.12: Inversion and Uniqueness of the Characteristic Function

If the probability measure μ has characteristic function ψ , and if $\mu\{a\} = \mu\{b\} = 0$, then

$$\mu(a, b] = \lim_{t \rightarrow \infty} \frac{1}{2\pi} \int_{\mathbb{R}} dr \mathbb{1}_{[-t, t]}(r) \frac{e^{-ira} - e^{-irb}}{ir} \psi(r) := \lim_{t \rightarrow \infty} I_t. \quad (2.11)$$

In particular, the characteristic function determines μ uniquely. If, in addition, $\int_{\mathbb{R}} dr |\psi(r)| < \infty$, then μ has a density f with respect to the Lebesgue measure, given by

$$f(x) := \frac{1}{2\pi} \int_{-\infty}^{\infty} dr e^{-irx} \psi(r). \quad (2.12)$$

Proposition 2.3: Properties of the Characteristic Function

- (1) (Symmetry): The random variables X and $-X$ are identically distributed if and only if the characteristic function of X is real-valued.
- (2) (Convolution): If X and Y are independent, then $\psi_{X+Y} = \psi_X \psi_Y$.
- (3) (Affine transformation): If a and b are real numbers, then

$$\psi_{aX+b}(r) = e^{irb} \psi_X(ar), \quad r \in \mathbb{R}.$$

- (4) (Taylor's theorem): If $\mathbb{E}|X^n| < \infty$, then

$$\psi(r) = \sum_{k=0}^n \frac{\mathbb{E}X^k}{k!} (ir)^k + O(r^n), \quad r \in \mathbb{R}.$$

2.7 Information and Independence

In this section, we have another look at the role of σ -algebras and independence in random experiments.

Definition 2.14: σ -Algebra Generated by a Random Variable

Let X be a random variable taking values in $(E, \mathcal{E})$. The sets $\{X \in A\}, A \in \mathcal{E}$ form a σ -algebra, called the σ -algebra generated by X , and we write σX .

Theorem 2.13: σ -Algebra Generated by a Stochastic Process

For the stochastic process $X := \{X_t, t \in \mathbb{T}\}$, the σ -algebra generated by X is the smallest σ -algebra on Ω that is generated by the union of the σ -algebras $\sigma X_t, t \in \mathbb{T}$. We write $\bigvee_{t \in \mathbb{T}} \sigma X_t$ or $\sigma\{X_t, t \in \mathbb{T}\}$.

Theorem 2.14: σX -Numerical Random Variables

Let X be a random variable taking values in a measurable space $(E, \mathcal{E})$. A mapping $V : \Omega \rightarrow \bar{\mathbb{R}}$ belongs to σX if and only if $V = h \circ X$ for some $h \in \mathcal{E}$.

Definition 2.15: Filtration

Let $\mathbb{T}$ be a subset of $\mathbb{R}$. A *filtration* is an increasing collection $\{\mathcal{F}_t : t \in \mathbb{T}\}$ of sub- σ -algebras of $\mathcal{H}$; that is, $\mathcal{F}_s \subseteq \mathcal{F}_t$ whenever $s < t$.

Theorem 2.15: Stochastic Process with Arbitrary Index Set

For each t in an arbitrary set $\mathbb{T}$, let X_t be a random variable taking values in a measurable set $(E_t, \mathcal{E}_t)$. Then, the mapping $V : \Omega \rightarrow \bar{\mathbb{R}}$ belongs to $\sigma\{X_t, t \in \mathbb{T}\}$ if and only if there exists a sequence (t_n) in $\mathbb{T}$ and a function h in $\bigotimes_n \mathcal{E}_{t_n}$ such that

$$V = h(X_{t_1}, X_{t_2}, \dots).$$

Definition 2.16: Independence for σ -Algebras

Let $\{\mathcal{F}_i, i = 1, \dots, n\}$ be a collection of sub- σ -algebras of $\mathcal{H}$. The $\{\mathcal{F}_i\}$ are said to be (mutually) *independent* or form an *independency* if

$$\mathbb{E}V_1 \cdots V_n = \mathbb{E}V_1 \cdots \mathbb{E}V_n \quad (2.13)$$

for all positive random variables $V_i \in \mathcal{F}_i, i = 1, \dots, n$.

Theorem 2.16: Independence of σ -Algebras

Sub- σ -algebras $\mathcal{F}_1, \dots, \mathcal{F}_n$ of $\mathcal{H}$ generated by p -systems $\mathcal{C}_1, \dots, \mathcal{C}_n$ are independent if and only if for all $H_i \in \mathcal{C}_i \cup \{\Omega\}$, $i = 1, \dots, n$, it holds that:

$$\mathbb{P}(H_1 \cap \dots \cap H_n) = \mathbb{P}(H_1) \dots \mathbb{P}(H_n). \quad (2.14)$$

Theorem 2.17: Independence of Random Variables

Random variables X and Y taking values in $(E, \mathcal{E})$ and $(F, \mathcal{F})$ are independent if and only if

$$\mathbb{E}g(X)h(Y) = \mathbb{E}g(X)\mathbb{E}h(Y), \quad g \in \mathcal{E}_+, h \in \mathcal{F}_+. \quad (2.15)$$

2.8 Important Stochastic Processes

We conclude by highlighting some key classes of stochastic processes, including Gaussian processes, Poisson random measures, and Lévy processes.

Definition 2.17: Gaussian Process

A real-valued stochastic process $\{X_t, t \in \mathbb{T}\}$ is said to be *Gaussian* if all its finite-dimensional distributions are Gaussian (normal); that is, if the vector $[X_{t_1}, \dots, X_{t_n}]^\top$ is multivariate normal for any choice of n and $t_1, \dots, t_n \in \mathbb{T}$.

Definition 2.18: Poisson Random Measure

A random measure N on $(E, \mathcal{E})$ is said to be a *Poisson random measure* with mean measure μ if the following properties hold:

- (1) $N(A) \sim \text{Poi}(\mu(A))$ for any set $A \in \mathcal{E}$.
- (2) For any selection of disjoint sets $A_1, \dots, A_n \in \mathcal{E}$, the random variables $N(A_1), \dots, N(A_n)$ are independent.

Proposition 2.4: Poisson Random Measure on a Product Space

Let $X := \{X_i, i \in I\}$ be the atoms of a Poisson random measure N on $(E, \mathcal{E})$ with mean measure μ , and let $\{Y_i, i \in \mathbb{N}\}$ be i.i.d. random variables on $(F, \mathcal{F})$ with distribution π , independent of X . Then, $\{(X_i, Y_i), i \in I\}$ form the atoms of a Poisson random measure M on $(E \times F, \mathcal{E} \otimes \mathcal{F})$ with mean measure $\mu \otimes \pi$.

Theorem 2.18: Conditioning on the Total Number of Points

Let N be a Poisson random measure on $(E, \mathcal{E})$ with mean measure μ satisfying $\mu(E) := c < \infty$. Then, conditional on $N(E) = k$, the k atoms of N are i.i.d. with probability distribution μ/c .

Definition 2.19: Compound Poisson Process

The process $(X_t, t \geq 0)$ defined by

$$X_t := \int_{[0,t] \times \mathbb{R}^d} N(ds, dx) x, \quad t \geq 0,$$

is the *compound Poisson process* corresponding to the measure ν .

Definition 2.20: Lévy Process

The process X is said to be a *Lévy process* if

- (1) the paths of X are almost surely right-continuous and left-limited, with $X_0 = 0$, and
- (2) for every $t_1 < t_2 < t_3 < t_4$, the increments $X_{t_2} - X_{t_1}$ and $X_{t_4} - X_{t_3}$ are independent, and for every $t, u \in \mathbb{R}_+$, the increment $X_{t+u} - X_t$ has the same distribution as X_u .

Theorem 2.19: Lévy-Itô Decomposition

The process $X := (X_t, t \geq 0)$ is a Lévy process if and only if there exists a d -dimensional Brownian motion process $B := (B_t, t \geq 0)$ starting at $\mathbf{0}$ and a Poisson random measure N on $\mathbb{R}_+ \times \mathbb{R}^d$ that is independent of B and has mean measure $\text{Leb} \otimes \nu$, with ν satisfying

$$\int_{\mathbb{R}^d} \nu(d\mathbf{x}) (|\mathbf{x}|^2 \wedge 1) < \infty, \quad (2.16)$$

such that $X = B + U + Z$, where the summands are independent, $U := (U_t, t \geq 0)$ is the compound Poisson process

$$U_t := \int_0^t \int_{\|\mathbf{x}\| > 1} N(ds, d\mathbf{x}) \mathbf{x}, \quad t \geq 0,$$

and $Z := (Z_t, t \geq 0)$ is the limit of compensated compound Poisson processes:

$$Z := \lim_{\delta \downarrow 0} Z^\delta \quad \text{with} \quad Z_t^\delta := \int_0^t \int_{\delta \leq \|\mathbf{x}\| \leq 1} [\mathbf{x} N(ds, d\mathbf{x}) - ds \mathbf{x} \nu(d\mathbf{x})], \quad t \geq 0.$$

3 Convergence

3.1 Motivation

Proposition 3.1: Cauchy Criteria for Convergence

The following statements satisfy (1) $\iff$ (2) and (3) $\implies$ (2):

- (1) (x_n) converges to some real $x := \lim x_n$.
- (2) (x_n) is Cauchy convergent, that is, $\lim_{m,n \rightarrow \infty} |x_m - x_n| = 0$.
- (3) $\sum_n \mathbb{1}_{\{|x_{n+1} - x_n| > \varepsilon_n\}} < \infty$ for any (ε_n) such that $c := \sum_n \varepsilon_n < \infty$.

3.2 Almost Sure Convergence

Definition 3.1: Almost Sure Convergence

The sequence of random variables (X_n) converges *almost surely* (a.s.) to a random variable X if

$$\mathbb{P}(\lim X_n = X) = 1.$$

We denote the almost sure convergence by $X_n \xrightarrow{\text{a.s.}} X$.

Theorem 3.1: Almost Sure Convergence

The sequence of random variables (X_n) converges almost surely to a random variable X if and only if for every $\varepsilon > 0$

$$\lim_{m \rightarrow \infty} \mathbb{P} \left(\sup_{n \geq m} |X_n - X| > \varepsilon \right) = 0. \quad (3.1)$$

Proposition 3.2: Cauchy Criteria for Almost Sure Convergence

The following statements are equivalent:

- (1) $X_n \xrightarrow{\text{a.s.}} X$.
- (2) $\lim_{n \rightarrow \infty} \mathbb{P}(\sup_{k \geq 0} |X_{n+k} - X_n| > \varepsilon) = 0$ for any choice of $\varepsilon > 0$.
- (3) $\lim_{k \rightarrow \infty} \mathbb{P}(\sup_{m, n \geq k} |X_m - X_n| > \varepsilon) = 0$ for any choice of $\varepsilon > 0$.
- (4) $\mathbb{P}(\lim_{m, n \rightarrow \infty} |X_m - X_n| = 0) = 1$; that is, (X_n) is a.s. a Cauchy sequence.

Proposition 3.3: Markov's and Chebyshev's Inequality

For any random variable Y and $\varepsilon > 0$, Markov's inequality holds:

$$\mathbb{P}(|Y| > \varepsilon) \leq \frac{\mathbb{E}|Y|}{\varepsilon}. \quad (3.2)$$

As a consequence, we have Chebyshev's inequality, where $\mu := \mathbb{E}X$:

$$\mathbb{P}(|X - \mu| > \varepsilon) \leq \frac{\text{Var } X}{\varepsilon^2}. \quad (3.3)$$

Theorem 3.2: Kolmogorov's Inequality

Let $S_n := \sum_{i=1}^n X_i$, where $X_1, X_2, \dots$ are independent with 0 mean. Then, for every $\varepsilon > 0$:

$$\mathbb{P}\left(\sup_{k \leq n} |S_k| > \varepsilon\right) \leq \frac{\text{Var } S_n}{\varepsilon^2}. \quad (3.4)$$

Lemma 3.1: Borel–Cantelli

Let (H_n) be a sequence of events. Then, the following hold:

- (1) $\sum_n \mathbb{P}(H_n) < \infty$ implies that $\mathbb{P}(\sum_n \mathbb{1}_{H_n} < \infty) = 1$.
- (2) $\sum_n \mathbb{P}(H_n) = \infty$ implies that $\mathbb{P}(\sum_n \mathbb{1}_{H_n} < \infty) = 0$, provided that the events (H_n) are pairwise independent.

3.3 Convergence in Probability**Definition 3.2: Convergence in Probability**

The sequence of random variables (X_n) converges in probability to a random variable X if, for all $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0.$$

We denote the convergence in probability by $X_n \xrightarrow{\mathbb{P}} X$.

Proposition 3.4: Convergence of a Subsequence

If (X_n) converges to X in probability, there is a subsequence (X_{n_k}) that converges to X almost surely.

Proposition 3.5: Cauchy Criterion for Convergence in Probability

The sequence of random variables (X_n) converges in probability if and only if for every $\varepsilon > 0$,

$$\lim_{m,n \rightarrow \infty} \mathbb{P}(|X_m - X_n| > \varepsilon) = 0. \quad (3.5)$$

3.4 Convergence in Distribution**Definition 3.3: Convergence in Distribution**

The sequence of random variables (X_n) is said to *converge in distribution* to a random variable X with cdf F provided that:

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n \leq x) = F(x) \quad \text{for all } x \text{ such that } \lim_{a \rightarrow x} F(a) = F(x). \quad (3.6)$$

We denote the convergence in distribution by $X_n \xrightarrow{d} X$.

Theorem 3.3: Characteristic Function and Convergence in Distribution

Suppose that $\psi_{X_1}(\mathbf{r}), \psi_{X_2}(\mathbf{r}), \dots$ are the characteristic functions of the sequence of d -dimensional random vectors $X_1, X_2, \dots$ and $\psi_X(\mathbf{r})$ is the characteristic function of X .

Then, the following three statements are equivalent:

- (1) $\lim_n \psi_{X_n}(\mathbf{r}) = \psi_X(\mathbf{r})$ for all $\mathbf{r} \in \mathbb{R}^d$.
- (2) $X_n \xrightarrow{d} X$.
- (3) $\lim_n \mathbb{E}h(X_n) = \mathbb{E}h(X)$ for all bounded continuous functions $h : \mathbb{R}^d \rightarrow \mathbb{R}$.

3.5 Convergence in L^p Norm

Definition 3.4: Convergence in L^p Norm

The sequence of random variables (X_n) converges in L^p norm (for some $p \in [1, \infty]$) to a random variable X , if $\|X_n\|_p < \infty$ for all n , $\|X\|_p < \infty$, and

$$\lim_{n \rightarrow \infty} \|X_n - X\|_p = 0.$$

We denote the convergence in L^p norm by $X_n \xrightarrow{L^p} X$.

Proposition 3.6: Cauchy Criterion for Convergence in L^p Norm

The sequence of random variables (X_n) converges in L^p norm for $p \in [1, \infty]$ if and only if $\|X_n\|_p < \infty$ for all n and

$$\lim_{m, n \rightarrow \infty} \|X_m - X_n\|_p = 0.$$

Proposition 3.7: Integrable Random Variable

A real-valued random variable X is integrable if and only if

$$\lim_{b \rightarrow \infty} \mathbb{E}|X| \mathbb{1}_{\{|X| > b\}} = 0. \quad (3.7)$$

Definition 3.5: Uniform Integrability

A collection $\mathcal{K}$ of random variables is said to be *uniformly integrable* (UI) if

$$\lim_{b \rightarrow \infty} \sup_{X \in \mathcal{K}} \mathbb{E}|X| \mathbb{1}_{\{|X| > b\}} = 0.$$

Proposition 3.8: Conditions for Uniform Integrability

Let $\mathcal{K}$ be a collection of random variables.

- (1) If $\mathcal{K}$ is a finite collection of integrable random variables, then it is UI.
- (2) If $|X| \leq Y$ for all $X \in \mathcal{K}$ and some integrable Y , then $\mathcal{K}$ is UI.
- (3) $\mathcal{K}$ is UI if and only if there is an increasing convex function f such that

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = \infty \quad \text{and} \quad \sup_{X \in \mathcal{K}} \mathbb{E}f(|X|) < \infty.$$

- (4) $\mathcal{K}$ is UI if $\sup_{X \in \mathcal{K}} \mathbb{E}|X|^{1+\varepsilon} < \infty$ for some $\varepsilon > 0$.
- (5) $\mathcal{K}$ is UI if and only if
 - (1) $\sup_{X \in \mathcal{K}} \mathbb{E}|X| < \infty$, and
 - (2) for every $\varepsilon > 0$ there is a $\delta > 0$ such that for every event H :

$$\mathbb{P}(H) \leq \delta \implies \sup_{X \in \mathcal{K}} \mathbb{E}|X| \mathbf{1}_H \leq \varepsilon. \quad (3.8)$$

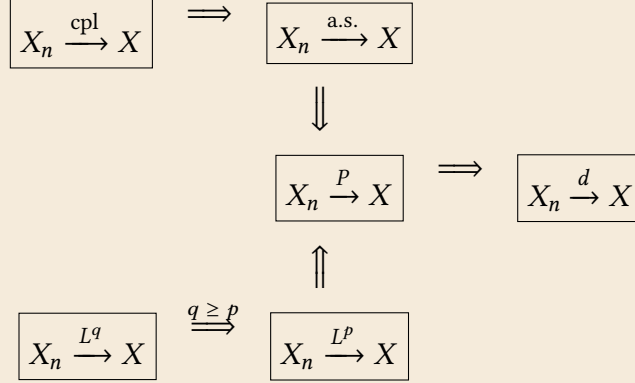
Theorem 3.4: L^1 Convergence and Uniform Integrability

A sequence (X_n) of real-valued integrable random variables converges in L^1 if and only if it converges in probability and is uniformly integrable.

3.6 Relations Between Modes of Convergence

Theorem 3.5: Modes of Convergence

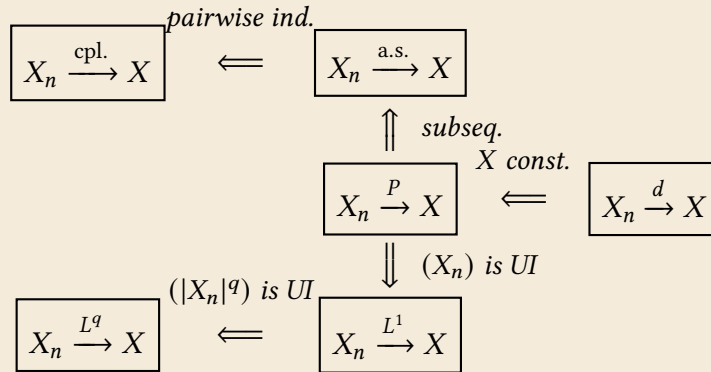
The most general relationships among the various modes of convergence for numerical random variables are shown below.



Under certain conditions, the converse statements can be established as follows:

Theorem 3.6: Modes of Convergence under Additional Assumptions

Converse relationships among the various modes of convergence are:



3.7 Law of Large Numbers and Central Limit Theorem

Theorem 3.7: Weak Law of Large Numbers

If $X_1, X_2, \dots$ are i.i.d. with finite expectation μ , then for all $\varepsilon > 0$

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) = 0.$$

In other words, $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$.

Theorem 3.8: Strong Law of Large Numbers

If $X_1, X_2, \dots$ are i.i.d. with expectation μ and $\mathbb{E}X^2 < \infty$, then for all $\varepsilon > 0$

$$\lim_{m \rightarrow \infty} \mathbb{P} \left(\sup_{n \geq m} |\bar{X}_n - \mu| > \varepsilon \right) = 0.$$

In other words, $\bar{X}_n \xrightarrow{a.s.} \mu$.

Theorem 3.9: Central Limit Theorem

If $X_1, X_2, \dots$ are i.i.d. with finite expectation μ and finite variance σ^2 , then for all $x \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq x \right) = \Phi(x),$$

where Φ is the cdf of the standard normal distribution. In other words, $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0, 1)$.

4 Conditioning

4.1 Conditional Expectation

In elementary probability, we define the conditional expectation after defining conditional probability. In advanced probability, it is more convenient to reverse the order, as we often want to condition on σ -algebras. In what follows, $(\Omega, \mathcal{H}, \mathbb{P})$ is a probability space, X a random variable, and $\mathcal{F}$ a sub- σ -algebra of $\mathcal{H}$; that is a σ -algebra whose every member (i.e., set) belongs to $\mathcal{H}$ as well. Recall the notation $V \in \mathcal{F}_+$ to mean that $V \geq 0$ and is $\mathcal{F}$ -measurable.

Definition 4.1: Conditional Expectation

The conditional expectation of X given $\mathcal{F}$, denoted $\mathbb{E}_{\mathcal{F}}X$, is defined in two steps:

- (1) For $X \geq 0$, $\mathbb{E}_{\mathcal{F}}X$ is defined as any random variable $\hat{X}$ with
 - (1) $\hat{X} \in \mathcal{F}_+$.
 - (2) $\mathbb{E}VX = \mathbb{E}V\hat{X}$ for every $V \in \mathcal{F}_+$ (*projection property*).
- (2) For arbitrary X , if $\mathbb{E}X$ exists (possibly $\pm\infty$), $\mathbb{E}_{\mathcal{F}}X := \mathbb{E}_{\mathcal{F}}X^+ - \mathbb{E}_{\mathcal{F}}X^-$. Otherwise, the conditional expectation is left undefined.

Note that the conditional expectation is defined *almost surely*.

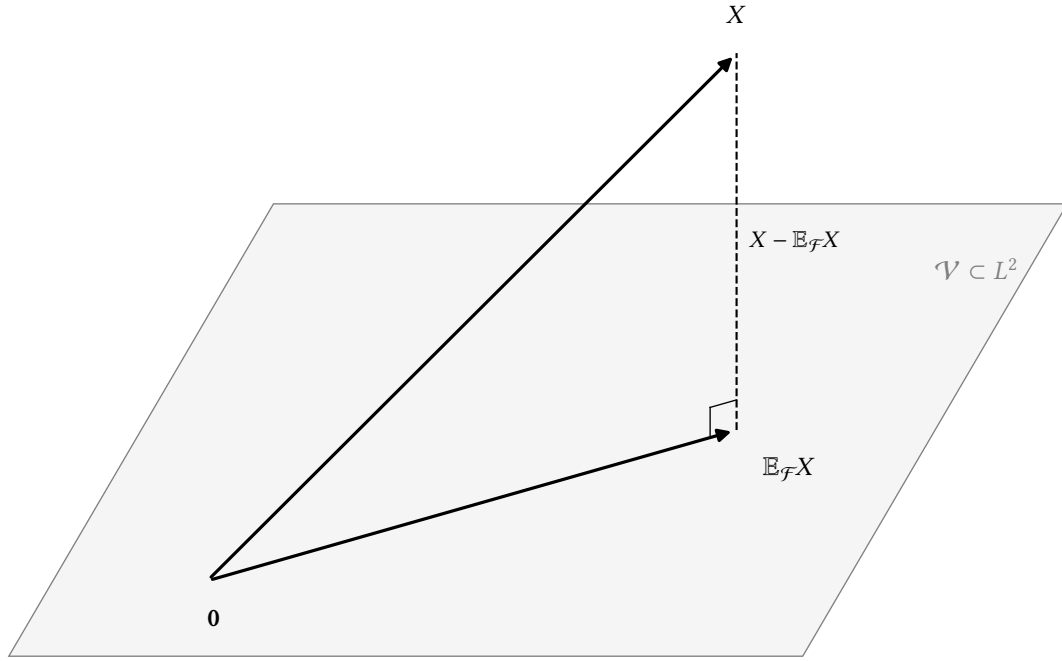


Figure 2: Geometric interpretation of Conditional Expectation $\mathbb{E}_{\mathcal{F}}X$ as an orthogonal projection.

Theorem 4.1: Existence and Uniqueness of the Conditional Expectation

Let $X \geq 0$ be a random variable and let $\mathcal{F}$ be a sub- σ -algebra of $\mathcal{H}$. Then, $\mathbb{E}_{\mathcal{F}}X$ exists and is unique up to equivalence.

Theorem 4.2: Additional Properties of the Conditional Expectation

Let $\mathcal{F}$ and $\mathcal{G}$ be sub- σ -algebras of $\mathcal{H}$. Let W and X be random variables such that $\mathbb{E}X$ and $\mathbb{E}WX$ exist. Then, the following hold:

- (1) (Taking out what is known): If $W \in \mathcal{F}$, then $\mathbb{E}_{\mathcal{F}}WX = W\mathbb{E}_{\mathcal{F}}X$.
- (2) (Repeated conditioning): If $\mathcal{F} \subseteq \mathcal{G}$, then

$$\mathbb{E}_{\mathcal{F}}\mathbb{E}_{\mathcal{G}}X = \mathbb{E}_{\mathcal{G}}\mathbb{E}_{\mathcal{F}}X = \mathbb{E}_{\mathcal{F}}X. \quad (4.1)$$

There is an amusing way to remember repeated conditioning: the σ -algebras $\mathcal{F}$ and $\mathcal{G}$ represent the information about the measurement on a random experiment by a fool and a

genius. The fool has no use for the genius' information,

$$\mathbb{E}_{\mathcal{F}}\mathbb{E}_{\mathcal{G}}X = \mathbb{E}_{\mathcal{F}}X,$$

and, sadly, the genius cannot further improve on the poor information of the fool,

$$\mathbb{E}_{\mathcal{G}}\mathbb{E}_{\mathcal{F}}X = \mathbb{E}_{\mathcal{F}}X.$$

In many applications of conditioning, we wish to take the expectation of a random variable X conditional on one or more random variables or, in general, on a stochastic process $Y := \{Y_t, t \in \mathbb{T}\}$, in which case $\mathcal{F} = \sigma(Y) = \sigma\{Y_t, t \in \mathbb{T}\}$. In this case we often write

$$\mathbb{E}_{\mathcal{F}}X = \mathbb{E}[X | Y_t, t \in \mathbb{T}] = \mathbb{E}[X | Y].$$

When conditioning on a random variable Y , we can use the fact (see [Theorem 2.14](#)) that any random variable in $\sigma(Y)$, in particular $\mathbb{E}[X | Y]$, must be a deterministic measurable function of Y ; that is, $\mathbb{E}[X | Y] = h(Y)$ for some measurable function h . This is in accordance with our use of conditional expectations in elementary probability: we first determine $h(y) := \mathbb{E}[X | Y = y]$ and then use $\mathbb{E}[X | Y] = h(Y)$ in computations.

Theorem 4.3: Conditional Expectation in L^2

For $X \in L^2$, the conditional expectation $\mathbb{E}_{\mathcal{F}}X$ is the orthogonal projection of X onto the subspace $\mathcal{V}$ of $\mathcal{F}$ -measurable random variables in L^2 .

4.2 Conditional Probability and Distribution

Definition 4.2: Conditional Probability for Events

Let A and B be two events with $\mathbb{P}(B) > 0$. The *conditional probability* that A occurs given that B occurs is denoted by

$$\mathbb{P}(A | B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \quad (4.2)$$

Theorem 4.4: Properties Involving Conditional Probabilities

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and $A, A_1, A_2, \dots, A_n, B_1, B_2, \dots$ events such that $\mathbb{P}(A_1 \cap \dots \cap A_n) > 0$ and $\{B_i\}$ form a partition of Ω , with $\mathbb{P}(B_i) > 0$ for all i . Then, the following hold:

(1) Law of total probability: $\mathbb{P}(A) = \sum_i \mathbb{P}(A | B_i) \mathbb{P}(B_i)$.

(2) Bayes' rule: $\mathbb{P}(B_j | A) = \frac{\mathbb{P}(B_j) \mathbb{P}(A | B_j)}{\sum_i \mathbb{P}(B_i) \mathbb{P}(A | B_i)}$.

(3) Chain or product rule:

$$\mathbb{P}(\cap_{i=1}^n A_i) = \mathbb{P}(A_1) \mathbb{P}(A_2 | A_1) \cdots \mathbb{P}(A_n | A_1 \cap \dots \cap A_{n-1}).$$

Definition 4.3: Conditional Distribution

Let Y be a random variable on the probability space $(\Omega, \mathcal{H}, \mathbb{P})$ taking values in the measurable space $(E, \mathcal{E})$. Let $\mathcal{F}$ be a sub- σ -algebra of $\mathcal{H}$. The *conditional distribution of Y given $\mathcal{F}$* is any probability transition kernel L from $(\Omega, \mathcal{F})$ to $(E, \mathcal{E})$ such that

$$\mathbb{P}_{\mathcal{F}}(Y \in B) = L(\cdot, B), \quad B \in \mathcal{E}. \quad (4.3)$$

Theorem 4.5: Existence of the Conditional Distribution

For every standard measurable space $(E, \mathcal{E})$ there exists a version of the conditional distribution of Y given $\mathcal{F}$.

Theorem 4.6: Regular Version of a Conditional Probability

Let $(\Omega, \mathcal{H})$ be a standard measurable space and $\mathcal{F}$ a sub- σ -algebra of $\mathcal{H}$. Then, there exists a regular version of the conditional probability $\mathbb{P}_{\mathcal{F}}$.

$$\pi(dx, dy) = \mu(dx) K(x, dy). \quad (4.4)$$

Theorem 4.7: Conditional Distribution

If π is of the form [eq. \(4.4\)](#), then the kernel L defined by

$$L(\omega, B) := K(X(\omega), B), \quad \omega \in \Omega, B \in \mathcal{E} \quad (4.5)$$

is a version of the conditional distribution of Y given X .

Theorem 4.8: Disintegration

Let π be a probability measure on the product space $(D \times E, \mathcal{D} \otimes \mathcal{E})$, where $(E, \mathcal{E})$ is a standard measurable space. Then, there exists a probability measure μ on $(D, \mathcal{D})$ and a transition kernel K from $(D, \mathcal{D})$ to $(E, \mathcal{E})$ such that

$$\pi(dx, dy) = \mu(dx)K(x, dy), \quad x \in D, y \in E. \quad (4.6)$$

Definition 4.4: Conditional Independence

Let $\mathcal{F}, \mathcal{F}_1, \dots, \mathcal{F}_n$ be sub- σ -algebras of $\mathcal{H}$. The $\{\mathcal{F}_i\}$ are said to be *conditionally independent* given $\mathcal{F}$ if

$$\mathbb{E}_{\mathcal{F}} V_1 \dots V_n = \mathbb{E}_{\mathcal{F}} V_1 \dots \mathbb{E}_{\mathcal{F}} V_n \quad (4.7)$$

for all positive random variables $V_1, \dots, V_n$ in $\mathcal{F}_1, \dots, \mathcal{F}_n$, respectively.

Proposition 4.1: Conditional Independence

$\mathcal{F}_1$ and $\mathcal{F}_2$ are conditionally independent given $\mathcal{F}$ if and only if

$$\mathbb{E}_{\mathcal{F} \vee \mathcal{F}_1} V_2 = \mathbb{E}_{\mathcal{F}} V_2 \quad \text{for all positive } V_2 \in \mathcal{F}_2. \quad (4.8)$$

$$\mathbb{P}(B_0 \times \dots \times B_n \times E_{n+1} \times \dots) = \pi_n(B_0 \times \dots \times B_n), \quad B_i \in \mathcal{E}_i, i = 0, \dots, n. \quad (4.9)$$

4.3 Existence of Probability Spaces**Theorem 4.9: Ionescu–Tulcea’s Existence Theorem**

There exists a unique probability measure $\mathbb{P}$ on $(\Omega, \mathcal{H})$ such that [eq. \(4.9\)](#) holds.

Theorem 4.10: Kolmogorov's Extension Theorem

Let $(E, \mathcal{E})$ be a standard measurable space and K an index set. For each finite subset J of K , let π_J be a probability measure on the product space $(E^J, \mathcal{E}^J)$. If the (π_J) are consistent in the sense that

$$\pi_I = \pi_J \circ \text{proj}_{JI}^{-1}, \quad I \subseteq J, \quad |J| < \infty,$$

then there exists a unique probability measure $\mathbb{P}$ on $(\Omega, \mathcal{H})$ and a stochastic process X taking values in $(E, \mathcal{E})^K$ such that, with $X_J := (X_j)_{j \in J}$:

$$\mathbb{P}(X_J \in A) = \pi_J(A), \quad A \in \mathcal{E}^J, \quad |J| < \infty.$$

Definition 4.5: Hölder Continuity

A function $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ is said to be *Hölder continuous* of order $\alpha > 0$ on $B \subseteq \mathbb{R}_+$ if there is a constant k such that

$$|f(t) - f(s)| \leq k|t - s|^\alpha, \quad s, t \in B.$$

Theorem 4.11: Hölder Continuous Modification

Let $X := (X_t, t \in [0, 1])$ be a stochastic process with state space $\mathbb{R}$. If there exist constants c, p, q in $(0, \infty)$ such that

$$\mathbb{E}|X_t - X_s|^p \leq c|t - s|^{1+q}, \quad s, t \in [0, 1], \quad (4.10)$$

then for every $\alpha \in (0, q/p)$ there is a modification $\tilde{X}$ of X that is almost surely Hölder continuous of order α on $[0, 1]$.

4.4 Markov Property

One of the most studied stochastic processes is the Markov process. In a way it can be viewed as the probabilistic analogue of a difference or differential equation, in the sense

that, conditional on the past history, future increments depend only on the present state. The concepts of conditional probability and expectation are thus essential to the analysis.

Throughout this section, $\mathbb{T}$ is some subset of $\mathbb{R}$ and $X := (X_t, t \in \mathbb{T})$ is a stochastic process on some probability space $(\Omega, \mathcal{H}, \mathbb{P})$, taking values in some measurable space $(E, \mathcal{E})$.

Definition 4.6: Adaptedness

A stochastic process $X := (X_t, t \in \mathbb{T})$ taking values in some measurable space $(E, \mathcal{E})$ is said to be *adapted* to a filtration $\mathcal{F} := (\mathcal{F}_t, t \in \mathbb{T})$ if for all $t \in \mathbb{T}$, X_t is $\mathcal{F}_t/\mathcal{E}$ -measurable.

In what follows, we assume that the process X is adapted to some filtration $\mathcal{F} := (\mathcal{F}_t, t \in \mathbb{T})$ – for example, the natural filtration of X , in which case $\mathcal{F}_t := \sigma(X_u, u \leq t, u \in \mathbb{T})$. Let $\mathcal{G}_t := \sigma(X_u, u \geq t, u \in \mathbb{T})$ be the *future* σ -algebra of X for time t .

Definition 4.7: Markov Property

X is said to be *Markovian* relative to $\mathcal{F}$ if for every t , the past $\mathcal{F}_t$ and the future $\mathcal{G}_t$ are conditionally independent, given the present state X_t .

Definition 4.8: Markovian Transition Function

A *Markovian transition function* on $(E, \mathcal{E})$ is a family $(P_{t,u}, t, u \in \mathbb{T}, t < u)$ of probability transition kernels on $(E, \mathcal{E})$ that satisfy the *Chapman–Kolmogorov equations*:

$$P_{s,t}P_{t,u} = P_{s,u}, \quad s < t \leq u. \quad (4.11)$$

A Markovian process X is said to admit $(P_{t,u})$ as a transition function if

$$\mathbb{E}[f(X_u) | X_t] = (P_{t,u}f)(X_t), \quad t < u, f \in \mathcal{E}_+.$$

For $f = 1_A$ with $A \in \mathcal{E}$, this means

$$\mathbb{P}(X_u \in A | X_t = x) = P_{t,u}(x, A),$$

giving an intuitive meaning to the transition functions.

Define $P_t := P_{0,t}$. If $P_{t,u} = P_{u-t}$ for all $t \leq u$, the associated Markovian process X is said to be *time-homogeneous* and P_t satisfies the *semi-group property*

$$P_{t+s} = P_s P_t, \quad s, t \geq 0.$$

Markov processes can be classified according to the nature (discrete or continuous) of their state space E and time set $\mathbb{T}$. We call a Markovian process a *Markov chain*¹ if $\mathbb{T}$ is discrete. The Markov process *par excellence* in continuous time is the Lévy process; this includes the Wiener, Poisson, compound Poisson, and pure-jump Lévy processes.

Theorem 4.12: Generator of a Markov Jump Process

The infinitesimal generator of a Markov jump process with Lévy kernel L is given by

$$(Gf)(x) = \int_{\mathbb{R}^d} L(x, dy) [f(y) - f(x)], \quad f \in \mathcal{B}^d(\text{bounded}). \quad (4.12)$$

Proposition 4.2: Poisson Integrals

Let N be a Poisson random measure on $\mathbb{R}_+ \times E$ with mean measure μ satisfying $\mu(\{0\} \times E) = 0$. Let $R := (R(t, z), t \geq 0, z \in E)$ be a positive stochastic process in $\mathcal{F}^P \otimes \mathcal{E}$. Then,

$$\mathbb{E}NR = \mathbb{E}\mu R. \quad (4.13)$$

¹There is no consensus on this nomenclature. Some call a Markovian process a Markov chain if E is discrete.

5 Martingales

Martingales are real-valued stochastic processes in continuous or discrete time that model “fair” betting games, in which at any time the expected future profit is always 0, irrespective of any information about the past of the game. Many proofs in probability theory are facilitated by the use of martingales. In the analysis of martingales, it will be useful to consider the process at certain random times, called *stopping times* (also referred to as *optional times*). We introduce these next.

5.1 Stopping Times

In what follows, $(\Omega, \mathcal{H}, \mathbb{P})$ is a probability space, $\mathbb{T}$ a subset of $\mathbb{R}$, and $\mathcal{F} := (\mathcal{F}_t, t \in \mathbb{T})$ a filtration of sub σ -algebras of $\mathcal{H}$. We add a point of infinity to $\mathbb{T}$ to define $\bar{\mathbb{T}} := \mathbb{T} \cup \{\infty\}$. Recall that each $\mathcal{F}_t$ is a sub σ -algebra of $\mathcal{H}$ that models all the information on a random experiment (described by the probability space) that is available at time t . If X describes our measurements on the random experiment, then the information we have available at time t is given by $\sigma(X_s, s \leq t)$. However, $\mathcal{F}_t$ could have more information than $\sigma(X_s, s \leq t)$; that is, $\mathcal{F}$ could be “finer” than the natural filtration of X .

Definition 5.1: Stopping Time

A random time $T : \Omega \rightarrow \bar{\mathbb{T}}$ is called a *stopping time* of $\mathcal{F}$ if

$$\{T \leq t\} \in \mathcal{F}_t \quad \text{for each } t \in \mathbb{T}.$$

Consider the stochastic process $(Z_t) := (\mathbb{1}_{\{T \leq t\}}, t \in \mathbb{T})$. Imagine that T is the time when a “catastrophe” happens in the random experiment, at which time the process (Z_t) jumps from 0 to 1. If $\mathcal{F}_t$ is the available information at time t , then T is a stopping time if and only if we can tell, at time t , using only the information available in $\mathcal{F}_t$, whether the catastrophe has occurred yet or not. In other words, for a stopping time T we are able to construct an alarm system that uses only the information in $\mathcal{F}$ and that for every $\omega \in \Omega$ sounds exactly at the time of catastrophe $T(\omega)$.

Definition 5.2: Past until T

Let $\mathcal{F}$ be a filtration on $\bar{\mathbb{T}}$, and let T be a stopping time of it. The σ -algebra of *past information until T* is defined as

$$\mathcal{F}_T := \{H \in \mathcal{H} : H \cap \{T \leq t\} \in \mathcal{F}_t \text{ for every } t \in \bar{\mathbb{T}}\}.$$

We can think of $\mathcal{F}_T$ as the information on a random experiment that is available at a random stopping time T . If T is constant, $T = t$, then $\mathcal{F}_T$ is simply $\mathcal{F}_t$. As usual, we will use the notation $\mathcal{F}_T$ also for the collection of numerical random variables that are $\mathcal{F}_T/\bar{\mathcal{B}}$ -measurable.

Theorem 5.1: $\mathcal{F}_T$ Random Variables

A random variable V belongs to $\mathcal{F}_T$ if and only if

$$X_t := V\mathbb{1}_{\{T \leq t\}} \in \mathcal{F}_t \text{ for every } t \in \bar{\mathbb{T}}.$$

Proof. Without loss of generality assume $V \geq 0$. For all $v \geq 0$ and $t \in \mathbb{T}$,

$$\{V > v\} \cap \{T \leq t\} = \{X_t > v\}.$$

Hence, $\{V > v\} \in \mathcal{F}_T$ for all $v \geq 0$ if and only if $X_t \in \mathcal{F}_t$ for every $t \in \bar{\mathbb{T}}$.

Intuitively, $V \in \mathcal{F}_T$ means that for every ω , the value $V(\omega)$ can be discovered at time $T(\omega)$. □

5.2 Martingales

We have already mentioned that a martingale can be thought of as a process modelling a fair game. We now make this more precise. Let $(X_t, t \in \mathbb{T})$ be a stochastic process describing the profit at each time t in a betting game ruled by chance. Suppose the information available at time t is given by the σ -algebra $\mathcal{F}_t$. If for any $s < t$ the expected incremental profit $X_t - X_s$ is 0 given $\mathcal{F}_s$, then the game is fair, in the sense that we cannot win whatever betting strategy we devise; the process is then a martingale.

As martingale analysis relies heavily on conditional expectations given a filtration $\mathcal{F} := (\mathcal{F}_t, t \in \mathbb{T})$, we adopt the notation

$$\mathbb{E}_s V = \mathbb{E}_{\mathcal{F}_s} V = \mathbb{E}[V \mid \mathcal{F}_s],$$

with $\mathbb{E}_s$ being the most economical and elegant. When a filtration is not specified, the natural filtration is assumed.

Definition 5.3: (Sub/Super)Martingale

Let $X := (X_t, t \in \mathbb{T})$ be a real-valued stochastic process that is adapted to a filtration $\mathcal{F}$, and for which each X_t is integrable. The process X is called an $\mathcal{F}$ -(sub/super)martingale if for all $t > s$:

$$\begin{aligned} \mathbb{E}_s[X_t - X_s] &\geq 0 \quad (\text{submartingale}), \\ \mathbb{E}_s[X_t - X_s] &= 0 \quad (\text{martingale}), \\ \mathbb{E}_s[X_t - X_s] &\leq 0 \quad (\text{supermartingale}). \end{aligned} \tag{5.1}$$

When specifying (sub/super)martingales, we will often drop the quantifier “ $\mathcal{F}$ ” if the corresponding filtration is obvious; for example, when it is the natural filtration of the process.

Using the betting game analogy, for submartingales the expected incremental profit is positive and for supermartingales it is negative. Think of a *sub*martingale as a process that starts *low* and has the tendency to *increase*. You want your betting game to be a submartingale! It suffices to only consider submartingales and martingales for further analysis, as the negative of a supermartingale is a submartingale. Note that in the definition we may replace $\mathbb{E}_s[X_t - X_s]$ with $\mathbb{E}_s X_t - X_s$, as $\mathbb{E}_s X_s = X_s$, by the “taking out what is known” property of the conditional expectation; see Theorem 4.4. From the same theorem we will frequently use the “repeated conditioning” property of the conditional expectation. In particular, taking the expectation of $\mathbb{E}_s X_t - X_s$ shows that a martingale has constant expectations:

$$\mathbb{E}X_t = \mathbb{E}X_s$$

Here are some more easy properties of (sub)martingales.

Theorem 5.2: Properties of (Sub)Martingales

- (1) When $\mathbb{T} = \mathbb{N}$, the martingale equality $\mathbb{E}_s[X_t - X_s] = 0$ holds if and only if $\mathbb{E}_k[X_{k+1} - X_k] = 0, k \in \mathbb{N}$.
- (2) If X and Y are $\mathcal{F}$ -submartingales, then so is $aX + bY$ for $a, b \geq 0$.
- (3) If X and Y are $\mathcal{F}$ -submartingales, then so is $X \vee Y$.
- (4) Let f be a convex function on $\mathbb{R}$ and X an $\mathcal{F}$ -martingale. If $f(X_t)$ is integrable for all t , then $f(X) := (f(X_t))$ is an $\mathcal{F}$ -submartingale.

5.3 Optional Stopping

The main result of this section is Doob's "optional" stopping theorem, which, under certain conditions, extends the martingale property, Equation (5.1) to stopping times $S < T$, to give

$$\mathbb{E}_S[X_T - X_S] = 0. \quad (5.2)$$

and consequently (assuming $0 \in \mathbb{T}$):

$$\mathbb{E}X_T = \mathbb{E}X_0. \quad (5.3)$$

This is certainly not true for *all* stopping times.

5.3.1 Stochastic Integration

Stochastic analysis often involves the evaluation of a stochastic integral; that is, an integral of the form

$$\int_0^t F_s dX_s, \quad (5.4)$$

where the Integrator $X \equiv (X_t)$ and Integrand $F \equiv (F_t)$ are real-valued stochastic processes on a probability space $(\Omega, \mathcal{H}, \mathbb{P})$. Let $f_t \equiv F_t(\omega)$ be the realization of F_t for a specific $\omega \in \Omega$, and similarly let $x_t \equiv X_t(\omega)$, with the additional assumption that $x \equiv (x_t)$ is right-continuous and left-limited. If, for each $\omega \in \Omega$, the real-valued function x is increasing and the function $f \equiv (f_t)$ is measurable, then the integral can be defined pathwise; that is, for each ω the integral

$$\int_0^t f_s dx_s \quad (5.5)$$

is well-defined. Since an increasing function x corresponds to a unique measure μ , we can define this integral as the Lebesgue integral

$$\int_0^t f_s \mu(ds). \quad (5.6)$$

The arguments above extend to the case when the function x can be written as the difference of two increasing functions, say $x = y - z$. Then the integral can again be defined pathwise as

$$\int_0^t f_s dy_s - \int_0^t f_s dz_s. \quad (5.7)$$

The corresponding pathwise integral is called the Lebesgue–Stieltjes Integral.

It is not difficult to show that such functions x are of bounded variation on $[0, t]$, meaning that their total variation on $[0, t]$ is finite. The latter is defined as

$$V_x(t) \equiv \sup_{(\Pi_n)} \sum_{k=0}^{n-1} |x_{s_{k+1}} - x_{s_k}|, \quad (5.8)$$

where the supremum is taken over any segmentation

$$\Pi_n \equiv \{s_0, \dots, s_n : 0 = s_0 < s_1 < \dots < s_n = t\}, \quad n \in \mathbb{N}, \quad (5.9)$$

such that the mesh of Π_n , defined as

$$\|\Pi_n\| \equiv \max_{0 \leq k \leq n-1} (s_{k+1} - s_k), \quad (5.10)$$

goes to 0 as $n \rightarrow \infty$. Any right-continuous function of bounded variation can be written as a difference of two increasing functions.

The upshot is that for a process X that does not wiggle too much, the stochastic integral can be defined pathwise as a Lebesgue-Stieltjes integral.

Definition 5.4: Predictable Process

Let $\mathcal{F} := (\mathcal{F}_t, t \geq 0)$ be a filtration. A real-valued stochastic process F is said to be $\mathcal{F}$ -predictable (or simply *predictable*) if it is measurable with respect to the σ -algebra

$$\mathcal{F}^P := \sigma(H \times (a, b] : a, b \in \mathbb{R}_+, H \in \mathcal{F}_a) \vee \sigma(H \times \{0\} : H \in \mathcal{F}_0). \quad (5.11)$$

Loosely speaking, this means that we are able to predict the value of F_t from all the information available before t .

For discrete processes $X := (X_n, n \in \mathbb{N})$ and $F := (F_n, n \in \mathbb{N})$, the treatment of stochastic integration becomes much easier. We simply define

$$\int_0^n F_k dX_k := F_0 X_0 + F_1 (X_1 - X_0) + \cdots + F_n (X_n - X_{n-1}), \quad n = 1, 2, \dots \quad (5.12)$$

For $n = 0$, the integral is defined as $F_0 X_0$. The justification is as follows. If X is an increasing process, then for any $\omega \in \Omega$, there corresponds to $X(\omega)$ a discrete measure with mass $m(k) := X_k(\omega) - X_{k-1}(\omega)$, at $k = 1, \dots, n$, and $m(0) := X_0(\omega)$. The corresponding Lebesgue integral with respect to $F(\omega)$ is then given by [Equation \(5.12\)](#), as a function of ω . If X is arbitrary, we can write X as the difference of two increasing processes and then take the integral as the difference between the positive and negative part. This is equivalent to associating with $X(\omega)$ a signed measure (i.e., the difference of two measures) with the same masses $m(k), k = 0, \dots, n$, given above. This thus defines the Lebesgue-Stieltjes integral in the discrete case; i.e., the Stieltjes integral interpreted as a Lebesgue integral.

Note that [Equation \(5.12\)](#) defines a new stochastic process

$$Z_n := \int_0^n F_k dX_k, \quad n \in \mathbb{N}. \quad (5.13)$$

Think of F_n as the number of shares owned at period n — say the time interval $(n-1, n]$ in years — and X_n as the price at the end of period n . Then, $(X_n - X_{n-1})F_n$ is the profit made during period n . Moreover, Z_n is the total capital at the end of period n , starting with an initial capital of X_0F_0 . If (X_n) is a martingale, then, by definition, conditional on the past until time $n-1$, the expected increase in share price is 0; that is, $\mathbb{E}_{n-1}(X_n - X_{n-1}) = 0$. We shall see that if F is a predictable process, then the expected profit during period n , given the information up to period $n-1$ is also 0. Here is the precise definition of predictability for the discrete case.

Definition 5.5: Discrete Predictable Process

Given a filtration $\mathcal{F} := (\mathcal{F}_n, n \in \mathbb{N})$, a process $F := (F_n, n \in \mathbb{N})$ is said to be *predictable* if $F_0 \in \mathcal{F}_0$ and $F_n \in \mathcal{F}_{n-1}$, $n = 1, 2, \dots$

Continuing the financial analogy, if the shares process F is predictable with respect to the natural filtration of the share price — that is, if the number of shares owned at period n is completely determined by the evolution of the share price in the preceding periods, then $\mathbb{E}_{n-1}(X_n - X_{n-1})F_n = F_n \mathbb{E}_{n-1}(X_n - X_{n-1}) = 0$, as reported. The total capital process (Z_n) defined in Equation (5.13) is in fact a martingale, as the following theorem shows:

Theorem 5.3: Integral Transform of a Martingale

Let F be a bounded predictable process and X a martingale adapted to a filtration $\mathcal{F}$. Then, the integral transform $Z_n := \int_0^n F_k dX_k$, $n \in \mathbb{N}$ is an $\mathcal{F}$ -martingale as well.

It is sometimes useful to consider a stochastic process that is stopped at some random time T .

Definition 5.6: Stopped Stochastic Process

Let $X := (X_t, t \in \mathbb{T})$ be a stochastic process and T a random time with values in $\overline{\mathbb{T}}$. By the process X *stopped at time T* we mean the process $(X_t^\dagger, t \in \mathbb{T})$ defined by

$$X_t^\dagger(\omega) := X_{T(\omega) \wedge t}(\omega) = \begin{cases} X_t(\omega) & \text{if } t \leq T(\omega), \\ X_{T(\omega)}(\omega) & \text{if } t > T(\omega). \end{cases} \quad (5.14)$$

Stopping a martingale at a random stopping time, gives again a martingale.

Theorem 5.4: Stopped Martingale

Let $M := (M_n, n \in \mathbb{N})$ be a martingale and T a stopping time with values in $\overline{\mathbb{N}}$. Then, the process M stopped at time T is again a martingale.

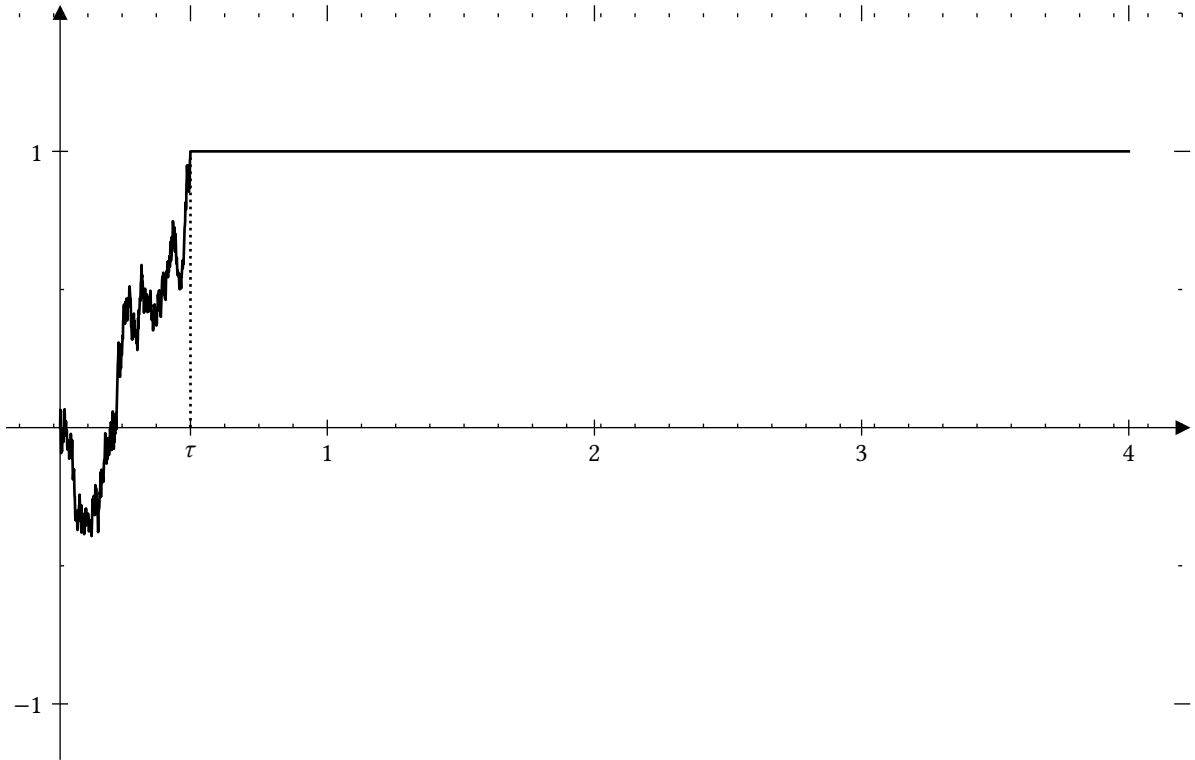


Figure 3: A Stopped Stochastic Process

Suppose that a martingale is stopped (frozen) if it will go up in the near future, but is allowed to continue if it will go down. Then, stopping introduces a downward bias by removing the upward possibility.

Proposition 5.1: Stochastic Integral for Predictable Integrands

Let N be defined as above. Then, for any positive $\mathcal{F}$ -predictable process (F_t) ,

$$\mathbb{E} \int_{\mathbb{R}_+} F_t dN_t = \mathbb{E} \int_{\mathbb{R}_+} F_t dv_t. \quad (5.15)$$

Theorem 5.5: Integral Transform of a Continuous Martingale

Let $\bar{N}$ be an $\mathcal{F}$ -martingale, as defined above. For any bounded $\mathcal{F}$ -predictable process F , the integral transform

$$Z_t := \int_0^t F_s d\bar{N}_s, \quad t \geq 0, \quad (5.16)$$

is a martingale.

5.3.2 Doob's Stopping Theorem**Theorem 5.6: Doob's Stopping Theorem**

A stochastic process $M := (M_n, n \in \mathbb{N})$ is a martingale if and only if for every pair of bounded stopping times S and T with $S \leq T$ the random variables M_S and M_T are integrable, and

$$\mathbb{E}_S(M_T - M_S) = 0. \quad (5.17)$$

Lemma 5.1: Maximum Inequality for Positive Submartingales

Let $(Y_k, k \in \mathbb{N})$ be a positive submartingale and define $Y_n^* := \max_{k \leq n} Y_k$. Then, for any $b > 0$,

$$b\mathbb{P}(Y_n^* \geq b) \leq \mathbb{E}[Y_n \mathbf{1}_{\{Y^* \geq b\}}] \leq \mathbb{E}Y_n. \quad (5.18)$$

Theorem 5.7: Doob–Kolmogorov Inequality

Let $M := (M_k)$ be a martingale in L^p for some $p \in [1, \infty)$. Then, for $b > 0$,

$$\mathbb{P}(\max_{k \leq n} |M_k| > b) \leq \frac{\mathbb{E}|M_n|^p}{b^p}. \quad (5.19)$$

5.4 (Sub) Martingale Convergence

The second major use of martingales is in the proof of convergence for stochastic processes. A first step in the analysis is to characterise the number of times a stochastic process “upcrosses” an interval.

5.4.1 Upcrossing

Let $X := (X_k, k \in \mathbb{N})$ be an adapted process with respect to some filtration and consider the integral transformation $Z_n := \int_0^n F_k dX_k$, $n \in \mathbb{N}$, where (F_k) is a predictable process. Continuing the financial analogy, think of X_k as the share price at time k and F_k as the number of shares owned at period k . It is completely determined by the share price history before period k . That is, the share buying strategy is determined by the information available before the k th period.

For example, the strategy could have the following rules: (1) we are allowed at most one share, (2) when the share price drops to level a or lower, we buy one share (if our portfolio is empty), and (3) when the share price reaches b or higher, for some fixed $b > a$, we sell the share, if we own one. The total profit at period n is $Z_n - Z_0$. If the share price at n (if any) is higher than what it was bought for, then the total profit $Z_n - Z_0$ is at least $(b - a)U_n$, where U_n is the number of times the process X has *upcrossed* the interval (a, b) during the periods $0, 1, \dots, n$.

As a special case, suppose that $X \geq 0$ is a *positive submartingale* and that $a = 0$. Thus, if at any time k when the price drops to 0, we buy a share (if we have no share), and we sell a share (if we own one) when the price hits b or any price higher. Again, we are allowed at most one share. We thus have

$$Z_n - Z_0 \geq b U_n, \quad (5.20)$$

where U_n is the number of upcrossings of $(0, b)$; i.e., the total number of shares sold up to period n .

Now observe that for each $k = 0, 1, 2, \dots$, we have:

$$\mathbb{E}_k[Z_{k+1} - Z_k] = \mathbb{E}_k[(X_{k+1} - X_k)F_{k+1}] = \underbrace{F_{k+1}}_{\leq 1} \underbrace{\mathbb{E}_k[X_{k+1} - X_k]}_{\geq 0} \leq \mathbb{E}_k[X_{k+1} - X_k]. \quad (5.21)$$

Summing the above terms from $k = 0$ to $n - 1$ and taking expectations gives

$$\mathbb{E}[Z_n - Z_0] \leq \mathbb{E}[X_n - X_0]. \quad (5.22)$$

Combining this with Equation (5.20) gives:

$$b \mathbb{E}U_n \leq \mathbb{E}[X_n - X_0]. \quad (5.23)$$

For a general interval (a, b) and a submartingale X , the above result is generalized as follows. Recall that for a random variable V , $V^+ := V \vee 0 = \max\{V, 0\}$.

Theorem 5.8: Expected Upcrossings of a Submartingale

Suppose that X is a submartingale. Then, the expected number of upcrossings $U_n(a, b)$ of (a, b) satisfies:

$$(b - a) \mathbb{E}U_n(a, b) \leq \mathbb{E}[(X_n - a)^+ - (X_0 - a)^+]. \quad (5.24)$$

Proof. The number of upcrossings of (a, b) for X is the same as the number of upcrossings of $(0, b - a)$ for the process $(X - a)^+$. The latter is a positive submartingale, for which, by Equation (5.23), we have Equation (5.24). $\square$

The upcrossing theorem will be useful for proving convergence results for (sub)martingales. Recall that submartingale has a tendency to increase. If this increase is constrained in some way, it is reasonable that the process would converge. The following theorem restrains the growth of the submartingale by means of a finiteness condition on the supremum of the $\mathbb{E}X_n^+$:

Theorem 5.9: (Sub)Martingale Convergence

Let $X := (X_n, n \in \mathbb{N})$ be a submartingale. Suppose that $\sup_n \mathbb{E}X_n^+ < \infty$. Then, X converges almost surely to an integrable random variable X_∞ .

Theorem 5.10: (Sub)Martingale Convergence and Uniform Integrability

Let $X := (X_n, n \in \mathbb{N})$ be a submartingale. Then, X converges almost surely and in L^1 if and only if it is uniformly integrable. Moreover, if X converges, setting $X_\infty := \lim X_n$ extends X to a submartingale $\bar{X} := (X_n, n \in \bar{\mathbb{N}})$.

Theorem 5.11: Uniformly Integrable Martingale

A process $M := (M_n, n \in \mathbb{N})$ is a UI martingale if and only if there exists an integrable random variable Z such that

$$M_n = \mathbb{E}_n Z, \quad n \in \mathbb{N}. \quad (5.25)$$

Moreover, then M converges almost surely and in L^1 to an integrable random variable

$$M_\infty := \mathbb{E}_\infty Z, \quad (5.26)$$

and $(M_n, n \in \overline{\mathbb{N}})$ is again a UI martingale. If $Z \in \mathcal{F}_\infty$, then $M_\infty = Z$.

Theorem 5.12: Convergence for Martingales in Reversed Time

For $\mathbb{T} = \{\dots, -2, -1, 0\}$, let $M := (M_n, n \in \mathbb{T})$ be a martingale with respect to the filtration $(\mathcal{F}_n, n \in \mathbb{T})$. Then, M is UI and, moreover, as $n \downarrow -\infty$ it converges almost surely and in L^1 to the integrable random variable $M_{-\infty} := \mathbb{E}_{-\infty} M_0$, where $\mathcal{F}_{-\infty} := \cap_{n \in \mathbb{T}} \mathcal{F}_n$.

Theorem 5.13: Stopping for Uniformly Integrable Martingales

If $M := (M_n, n \in \overline{\mathbb{N}})$ is a uniformly integrable martingale, then for every pair of stopping times S and T with $S \leq T$ it holds that

$$\mathbb{E}_S M_T = M_S$$

In particular,

$$\mathbb{E} M_T = \mathbb{E} M_0.$$

5.5 Applications

5.5.1 Kolmogorov's 0–1 Law

Theorem 5.14: Kolmogorov's 0–1 Law

If $H \in \mathcal{T}$, then $\mathbb{P}(H)$ is either 0 or 1. Consequently, a numerical random variable Z that is $\mathcal{T}$ -measurable must be almost surely constant.

5.5.2 Radon-Nikodym Theorem

Definition 5.7: Separable σ -Algebra

A σ -algebra on Ω is said to be *separable* if it is generated by a sequence (H_n) of subsets of Ω .

Theorem 5.15: Radon-Nikodym

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space, where $\mathcal{H}$ is separable, and let Q be a finite measure on $(\Omega, \mathcal{H})$ with $Q \ll \mathbb{P}$. Then, there exists a positive random variable $Z \in \mathcal{H}$, written $Z = dQ/d\mathbb{P}$, such that $Q = Z\mathbb{P}$; that is,

$$Q(H) = \int_H Z d\mathbb{P} = \mathbb{E}1_H Z, \quad H \in \mathcal{H}. \quad (5.27)$$

Definition 5.8: Right-continuous Filtration

A filtration $\mathcal{F} := (\mathcal{F}_t)$ is said to be *right-continuous* if for all t ,

$$\bigcap_{\varepsilon > 0} \mathcal{F}_{t+\varepsilon} = \mathcal{F}_t.$$

Definition 5.9: Local Martingale

A process $(Z_t, t \geq 0)$ is called a *local martingale* if there exists a sequence of stopping times $(T_n, n \in \mathbb{N})$, called a *localizing sequence*, such that $T_n \rightarrow \infty$ almost surely and each stopped process $(Z_{t \wedge T_n}, t \geq 0)$ is a martingale.

Theorem 5.16: Properties of Local Martingales

Let $Z := (Z_t, t \geq 0)$ be a local martingale with localizing sequence $(T_n, n \in \mathbb{N})$. Then, the following hold:

- (1) If $Z \geq 0$ and $\mathbb{E}Z_0 < \infty$, then Z is a supermartingale.
- (2) If for each $t \geq 0$ the sequence $(Z_{t \wedge T_n}, n \in \mathbb{N})$ is uniformly integrable, then Z is a martingale.

Definition 5.10: Doob Martingale

Let ζ be a stopping time. A process $M := (M_t, t \in \mathbb{T})$ is called a *Doob martingale* on $[0, \zeta]$ if for all stopping times $0 \leq S \leq T \leq \zeta$ it holds that

$$\mathbb{E}_S M_T = M_S, \quad (5.28)$$

where M_S and M_T are integrable.

Theorem 5.17: Characterization of a Doob Martingale

Let ζ be a stopping time. A process $M := (M_t, t \in \mathbb{T})$ is a Doob martingale on $[0, \zeta]$ if and only if for every stopping time $T \leq \zeta$,

$$\mathbb{E}M_T = \mathbb{E}M_0. \quad (5.29)$$

Theorem 5.18: Doob's Stopping Theorem for Continuous Martingales

A martingale $M := (M_t, t \in \mathbb{R}_+)$ is a Doob martingale on $[0, b]$ for every $b \in \mathbb{R}_+$ and satisfies

$$\mathbb{E}_S M_T = M_S \quad (5.30)$$

for every pair (S, T) of bounded stopping times with $S \leq T$.

Proposition 5.2: Doob's Maximal Inequality

Let $X := (X_t, t \geq 0)$ be a submartingale that is positive and continuous. Then, for $p \geq 1$ and $b > 0$,

$$\mathbb{P}(\max_{0 \leq s \leq t} X_s \geq b) \leq \frac{\mathbb{E}X_t^p}{b^p}. \quad (5.31)$$

Proposition 5.3: Doob's Norm Inequality

Let $M := (M_t, t \geq 0)$ be a continuous martingale in L^p for some $p > 1$ and let $Z_t := \max_{s \leq t} |M_s|$. Then,

$$\mathbb{E}Z_t^p \leq q^p \mathbb{E}|M_t|^p, \text{ with } q := p/(p-1). \quad (5.32)$$

Lemma 5.2: Continuous Martingales with Finite Variation

Let $X := (X_t, t \geq 0)$ be a martingale with continuous paths and total variation $V_t < \infty$ on each interval $[0, t]$. Then, X is almost surely constant.

Proposition 5.4: Doob Martingale on $\overline{\mathbb{R}}_+$

A process M is a Doob martingale on $\overline{\mathbb{R}}_+$ if and only if it is a martingale on $\overline{\mathbb{R}}_+$. If so, then

$$\mathbb{E}_S M_T = M_S \quad (5.33)$$

for arbitrary stopping times with $S \leq T$.

Theorem 5.19: Extension of UI Martingales

A martingale M on $\mathbb{R}_+$ can be extended to a Doob martingale $\overline{M}$ on $\overline{\mathbb{R}}_+$ if and only if it is uniformly integrable; that is, if and only if

$$M_t = \mathbb{E}_t Z, \quad t \in \mathbb{R}_+ \quad (5.34)$$

for some integrable random variable Z . Moreover, then, it converges almost surely and in L^1 to an integrable random variable M_∞ and can be extended to a UI martingale on $\overline{\mathbb{R}}_+$.

Proposition 5.5: Sufficient Condition for Being Doob

Let M be a martingale on $\mathbb{R}_+$ and ζ a stopping time. If, almost surely,

$$\sup_{t \in [0, \zeta] \cap \mathbb{R}_+} |M_t| \leq Z, \quad (5.35)$$

where $\mathbb{E}Z < \infty$ (i.e., M is dominated by an integrable random variable Z), then $M_\zeta = \lim M_{\zeta \wedge t}$ exists and is integrable. Moreover, M is a Doob martingale on $[0, \zeta]$.

6 Wiener and Brownian Motion Processes

6.1 Wiener Process

We defined the Wiener process as a Gaussian process as follows:

Definition 6.1: Wiener Process

The *Wiener process* $(W_t, t \geq 0)$ is a zero-mean Gaussian process with continuous sample paths and covariance function

$$\gamma_{s,t} = \min\{s, t\} =: s \wedge t, \quad s, t \geq 0.$$

A Wiener process is a Lévy process with *continuous* sample paths, as shown in the following result, which can also serve as an equivalent definition of the Wiener process.

Theorem 6.1: Wiener Process as a Lévy Process

$W = (W_t)_{t \geq 0}$ is a Wiener process if and only if the following three properties hold:

- (1) (*Independent increments*): For any $t_1 < t_2 \leq t_3 < t_4$, the increments $W_{t_4} - W_{t_3}$ and $W_{t_2} - W_{t_1}$ are independent.
- (2) (*Gaussian stationarity*): For all $t, u \geq 0$, $W_{t+u} - W_t \sim \mathcal{N}(0, u)$.
- (3) (*Continuity of paths*): W has continuous paths, with $W_0 = 0$.

The following algorithm, [algorithm 1](#) uses the same ideas as in the previous proof to simulate the Wiener process at specific time points $t_1, \dots, t_n$

Algorithm 1 Simulating a Wiener Process

- 1: Let $0 = t_0 < t_1 < \dots < t_n$ be the times at which the process needs to be simulated.
 - 2: Simulate $Z_1, \dots, Z_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$.
 - 3: Return $W_0 := 0$ and $W_{t_k} := \sum_{i=1}^k Z_i \sqrt{t_i - t_{i-1}}, \quad k = 1, \dots, n$.
-

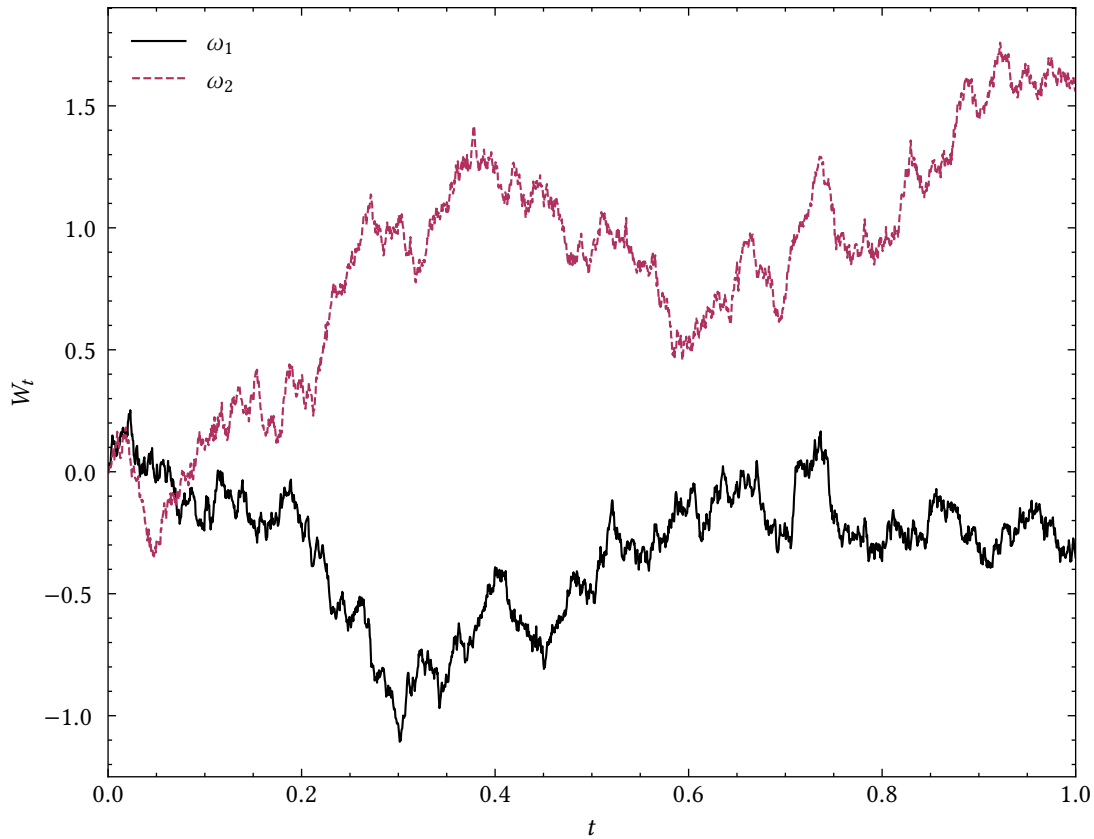


Figure 4: Two realisations of the Wiener process on the interval $[0, 1]$.

Definition 6.2: Brownian Motion

A stochastic process $(B_t, t \geq 0)$ satisfying

$$B_t = B_0 + \mu t + \sigma W_t, \quad t \geq 0,$$

where (W_t) is a Wiener process independent of B_0 , is called a *Brownian motion with drift μ and diffusion coefficient σ* .

A *standard Brownian motion* is one where $\mu = 0$ and $\sigma = 1$. The only difference between a Wiener process and a Brownian motion is its (random) starting position.

Algorithm 2 Simulating a Brownian Motion

- 1: Simulate the starting position B_0 .
 - 2: Simulate outcomes $W_{t_1}, \dots, W_{t_n}$ of a Wiener process at times $t_1, \dots, t_n$.
 - 3: Return $B_{t_i} := B_0 + \mu t_i + \sigma W_{t_i}$, $i = 1, \dots, n$ as the outcomes of the Brownian motion at times $t_1, \dots, t_n$.
-

Multidimensional Brownian motions are likewise obtained from an affine transformation $B_t = B_0 + \mu t + \sigma W_t$ of a multidimensional Wiener process, where σ is a matrix.

Definition 6.3: Multidimensional Wiener Process

Let $(W_{t,i}, t \geq 0)$, $i = 1, \dots, d$ be independent Wiener processes and let $W_t := [W_{t,1}, \dots, W_{t,d}]^\top$. The process $(W_t, t \geq 0)$ is called a d -dimensional Wiener process.

If (W_t) is a d -dimensional Wiener process, then W_t has a $\mathcal{N}(\mathbf{0}, t\mathbf{I}_d)$ multivariate normal distribution, where $\mathbf{I}_d$ denotes the d -dimensional identity matrix.

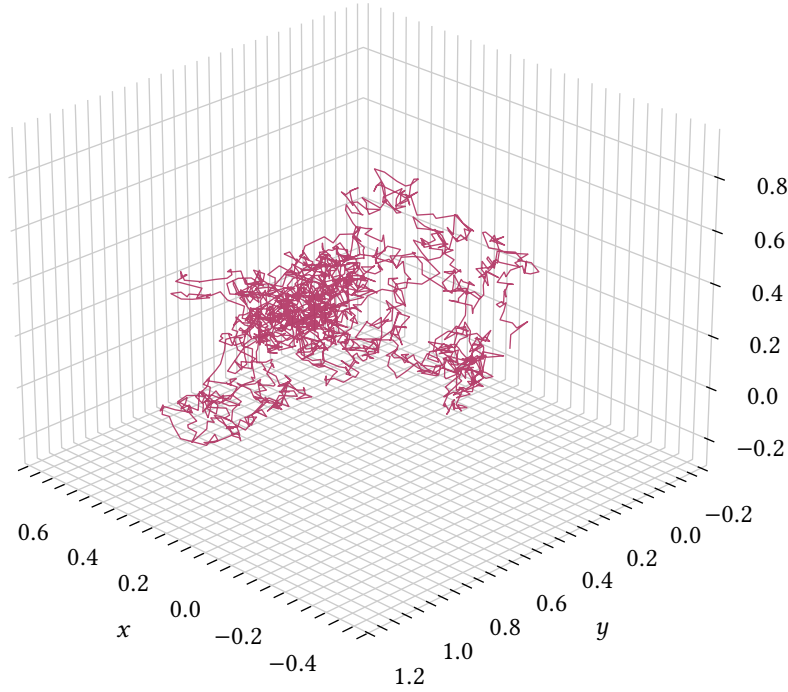


Figure 5: Three-dimensional Wiener process.

Theorem 6.2: Time-reversal and Time-shifting

If $(W_s, s \in [0, t])$ is a Wiener process on $[0, t]$, then so is the time-reversed process $(\widehat{W}_s, s \in [0, t])$ defined by $\widehat{W}_s := W_{t-s} - W_t$. Similarly, if $(W_s, s \geq 0)$ is a Wiener process, then for any $t \geq 0$ the process $(\widetilde{W}_s, s \geq 0)$ defined by $\widetilde{W}_s := W_{t+s} - W_t$ is also a Wiener process.

Proof. Obviously, $(\widehat{W}_s, s \in [0, t])$ is a Gaussian process with zero mean and continuous sample paths, as these properties are inherited from W . It remains to check that $\mathbb{E}[\widehat{W}_s \widehat{W}_u] =$

s for all $0 < s < u < t$. This follows from

$$\begin{aligned}
 \mathbb{E}[\widehat{W}_s \widehat{W}_u] &= \mathbb{E}[(W_{t-s} - W_t)(W_{t-u} - W_t)] \\
 &= \mathbb{E}[W_{t-s}W_{t-u}] + \mathbb{E}[W_t^2] - \mathbb{E}[W_t W_{t-u}] - \mathbb{E}[W_{t-s}W_t] \\
 &= (t-u) + t - (t-u) - (t-s) \\
 &= s.
 \end{aligned}$$

A similar argument shows that $(\widetilde{W}_s, s \geq 0)$ is a Wiener process. $\square$

Theorem 6.3: Scaling

If (W_t) is a Wiener process, then so is (X_t) , with $X_t := W_{at}/\sqrt{a}$, $t \geq 0$ for any $a > 0$.

Proof. That the scaled process is continuous, Gaussian, and has zero-mean is obvious. For $s < t$, the covariance function satisfies

$$\mathbb{E}X_s X_t = \mathbb{E} \frac{W_{as}}{\sqrt{a}} \frac{W_{at}}{\sqrt{a}} = \frac{as}{a} = s,$$

and so is the same as for the Wiener process. $\square$

Lemma 6.1: Probability Bound on the Supremum

For a Wiener process (W_t) , it holds for every $\varepsilon > 0$ that

$$\mathbb{P} \left(\sup_{0 \leq t \leq 1} |W_t| > \varepsilon \right) \leq 2e^{-\varepsilon^2/2}.$$

Theorem 6.4: Reciprocal Time

If (W_t) is a Wiener process, then so is (X_t) , with $X_t := tW_{1/t}$, $t > 0$, $X_0 := 0$.

Proof. Evidently, $(X_t, t > 0)$ is a zero-mean Gaussian process with continuous paths and with $\mathbb{E}X_s X_t = s$ for all $0 < s < t$. The only thing to check is that $(X_t, t \geq 0)$ —that is, including $t = 0$ —has the same properties; in particular, that the process is almost surely continuous at 0. In other words, we need to show that as $t \rightarrow \infty$,

$$\frac{W_t}{t} \xrightarrow{\text{a.s.}} 0. \tag{6.1}$$

Take any $t > 0$ and let $n := \lfloor t \rfloor$. Then, we can write

$$W_t = \sum_{k=1}^n Z_k + (W_t - W_n),$$

where the increments $Z_k := W_k - W_{k-1} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ are independent of $W_t - W_n$. This leads to the bound

$$\left| \frac{W_t}{t} \right| \leq \frac{1}{n} \left| \sum_{k=1}^n Z_k + (W_t - W_n) \right| \leq \left| \frac{1}{n} \sum_{k=1}^n Z_k \right| + \frac{1}{n} \sup_{0 \leq s \leq 1} |\tilde{W}_s|,$$

where $\tilde{W}_s := W_{n+s} - W_n$. By the Strong Law of Large Numbers, [Theorem 3.8](#) $\sum_{k=1}^n Z_k/n$ converges almost surely to 0. Since $(\tilde{W}_s, s \geq 0)$ is a Wiener process by [Theorem 6.2](#), we have by [Lemma 6.1](#) that for any $\varepsilon > 0$,

$$\mathbb{P} \left(\frac{1}{n} \sup_{0 \leq s \leq 1} |\tilde{W}_s| > \varepsilon \right) \leq 2e^{-n^2 \varepsilon^2 / 2}.$$

Since $\sum_n e^{-n^2 \varepsilon^2 / 2} < \infty$, it follows that $\frac{1}{n} \sup_{0 \leq s \leq 1} |\tilde{W}_s| \xrightarrow{\text{cpl}} 0$. Hence, [Theorem 3.5](#) implies that $\frac{1}{n} \sup_{0 \leq s \leq 1} |\tilde{W}_s| \xrightarrow{\text{a.s.}} 0$ and consequently that [Equation \(6.1\)](#) holds. $\square$

Tip 6.1: Starting Position



Since the Wiener process starts at position 0 by definition, it is sometimes useful to consider instead a standard Brownian motion process starting from some arbitrary state x under a probability measure $\mathbb{P}^x$. The corresponding expectation operator is then denoted by $\mathbb{E}^x$.

6.2 Existence

Theorem 6.5: Existence of the Wiener Process

There exists a probability space $(\Omega, \mathcal{H}, \mathbb{P})$ and a zero-mean Gaussian process $W := (W_t, t \in [0, 1])$ with covariance function $\mathbb{E}W_s W_t = s \wedge t$ and uniformly continuous sample paths.

6.3 Strong Markov Property

Theorem 6.6: Right-continuous Filtration

For the Wiener process, the augmentation $(\overline{\mathcal{F}}_t)$ of the natural filtration is right-continuous; that is,

$$\overline{\mathcal{F}}_t = \overline{\mathcal{F}}_t^+ := \bigcap_{\varepsilon > 0} \overline{\mathcal{F}}_{t+\varepsilon}.$$

Theorem 6.7: Wiener Process is Markov w.r.t $\mathcal{F}^+$

Let W be a Wiener process. For every $t \geq 0$, the process $(W_{t+u} - W_t, u \geq 0)$ is independent of $\mathcal{F}_t^+$. Consequently, W is Markovian with respect to $\mathcal{F}^+$.

Theorem 6.8: Strong Markov Property

For any finite stopping time T with respect to $\mathcal{F}^+$, the process $(W_{T+u} - W_T, u \geq 0)$ is a Wiener process independent of $\mathcal{F}_T^+$.

Theorem 6.9: Reflection Principle

Let T be a stopping time. Then, $(\tilde{W}_t, t \geq 0)$, defined by

$$\tilde{W}_t := W_t \mathbb{1}_{\{t \leq T\}} + (2W_T - W_t) \mathbb{1}_{\{t > T\}}, \quad t \geq 0, \quad (6.2)$$

is a Wiener process.

Theorem 6.10: Transition Kernel of the Wiener Process

The Wiener process W admits the transition function

$$P_{t,u}(x, A) = \mathbb{P}(W_u \in A \mid W_t = x) = \int_A dy p_{u-t}(y \mid x),$$

where the transition density $p_t(y \mid x)$ is given by

$$p_t(y \mid x) := \frac{1}{\sqrt{2\pi t}} e^{-\frac{(y-x)^2}{2t}}, \quad t \geq 0, \quad x, y \in \mathbb{R}. \quad (6.3)$$

Theorem 6.11: Exponential Martingale

A continuous process $X := (X_t, t \geq 0)$ is a Wiener process if and only if for each $r \in \mathbb{R}$ the process $S := (S_t)$ defined by

$$S_t := e^{rX_t - \frac{1}{2}r^2t}, \quad t \geq 0, \quad (6.4)$$

is a martingale.

Theorem 6.12: Martingales from Functions of a Brownian Motion

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be twice continuously differentiable and let B be d -dimensional standard Brownian motion. If for all $t > 0$ and $x \in \mathbb{R}^d$,

$$\mathbb{E}^x |f(B_t)| < \infty \quad \text{and} \quad \mathbb{E}^x \int_0^t |\Delta f(B_s)| ds < \infty,$$

then the process $(X_t, t \geq 0)$ defined by

$$X_t := f(B_t) - \frac{1}{2} \int_0^t \Delta f(B_s) ds \quad (6.5)$$

is a martingale.

7 Itô Calculus

Part II

Levy Processes

Part III

Stochastic Volatility

Part IV

Risk-Neutral Valuation

8 Discrete-time Market Model

9 Continuous-time Market Model

Part V

Martingale Methods

10 The Black-Scholes Model

11 A Dividend-paying Stock

12 Discretisation Schemes

13 Connection to Partial Differential Equations

Part VI

Monte Carlo Methods

SIMULATION techniques are invaluable in managing complex financial scenarios that involve path-dependent payoffs or distributions without analytical expressions. One prominent application is pricing derivative securities, estimating value at risk, and emulating hedging strategies. Monte Carlo simulation, in particular, proves highly effective for such tasks, offering a blend of simplicity and adaptability as its primary advantages. However, the method's drawback lies in its considerable computational demands, which can be mitigated to some extent. Our approach employs Monte Carlo simulation to generate simulated option price values, which are then compared against outcomes derived from analytical formulas.

14 Generating Random Numbers and Random Variables

15 Quasi Monte Carlo

IT is customary in this setting to focus on the problem of numerical integration over the unit hypercube.

The digital constructions are similar to feedback shift register random number generators, while lattice rules are similar to congruential generators. One useful way to think of QMC is that we are utilising a small random number generator in its entirety. The tools used to develop and analyse low-discrepancy methods are very different from those used in ordinary Monte Carlo, as they draw on number theory and abstract algebra rather than probability and statistics.

15.1 Discrepancy Measures

15.2 Low-discrepancy Sequences

15.2.1 Halton and Hammersley

15.2.2 Faure

15.2.3 Sobol

15.3 Koksma-Hlawka

16 Variance Reduction

Part VII

Grid & Lattice Methods

17 Lattice Methods

18 Finite Difference Methods

19 Contract Sensitivities

19.1 Likelihood Ratio

19.2 Pathwise Derivatives

19.3 Finite-Difference Approximations

Part VIII

Financial Derivatives

20 European Options

21 American Options

European option contracts specify an expiration date, and if the option is to be exercised at all, the exercise must occur on the expiration date. An option whose owner can choose to exercise at any time up to and including the expiration date is called *American*.

Whereas a European option can be exercised only at a fixed date, an American option can be exercised any time up to its expiration. The value of an American option is the value achieved by exercising optimally. Finding this value entails finding the optimal exercise rule by solving an optimal stopping problem and computing the expected discounted payoff of the option under this rule. The embedded optimisation problem makes this a difficult problem for simulation.

Because an American option can be exercised at any time prior to its expiration, it can never be worth less than the payoff associated with immediate exercise. This is called the *intrinsic value* of the option.

To price an American option, just as with a European option, we could imagine selling the option in exchange for some initial capital and then consider how to use this capital to hedge the short position in the option. In this case, we would need to be ready to pay off the option at all times prior to the expiration date because we do not know when it will be exercised. We could determine when, from our point of view, is the worst time for the owner to exercise the option. From the owner's point of view, this would be the *optimal exercise time*, and we shall call it that. We could then compute the initial capital we need in order to be hedged against exercise at the optimal exercise time. Finally, we could show how to invest this capital so that we are hedged even if the owner exercises at a nonoptimal time. In the subsequent sections, we do all these things but begin the analysis at a different point than for European options. We define the price of American options using a risk-neutral pricing formula and then show that this price is the smallest initial capital that permits construction of the hedge just described.

21.1 Perpetual American Put

The simplest interesting American option is the *perpetual American put*. It is interesting because the optimal exercise policy is not obvious, and it is simple because this policy can be determined explicitly. Although this is not a traded option, we begin our discussion with it in order to present in a simple context the ideas behind the subsequent analysis of more realistic options. The underlying asset (except where the asset pays dividends, which will be explicitly mentioned) has the price process $S(t)$ given by

$$dS(t) = rS(t), dt + \sigma S(t), d\tilde{W}(t), \quad (21.1)$$

where the interest rate r and the volatility σ are strictly positive constants and $\tilde{W}(t)$ is a Brownian motion under the risk-neutral probability measure $\tilde{\mathbb{P}}$. The perpetual American put pays $K - S(t)$ if it is exercised at time t . This is its *intrinsic value*.

Definition 21.1: Perpetual American Put

Let $\mathcal{T}$ be the set of all stopping times. The price of the perpetual American put is defined to be

$$v_*(x) = \max_{\tau \in \mathcal{T}} \tilde{\mathbb{E}} [e^{-r\tau} (K - S(\tau))], \quad (21.2)$$

where $x = S(0)$ in Equation (21.2) is the initial stock price. In the event that $\tau = \infty$, we interpret $e^{-r\tau} (K - S(\tau))$ to be zero.

The idea behind Definition 21.1 is that the owner of the perpetual American put can choose an exercise time τ , subject only to the condition that she may not look ahead to determine when to exercise. The mathematical formulation of this “not look ahead” restriction is that τ must be a stopping time. The price of the option at time zero is the risk-neutral expected payoff of the option, discounted from the exercise time back to time zero. If the option is never exercised, its payoff is zero. This explains the term under the expectation on the right-hand side of Equation (21.2). The owner of the option should choose the exercise strategy that maximizes this expected payoff, discounted back to time zero, and thus we define the price of the option to be the maximum over $\tau \in \mathcal{T}$ of the discounted expected payoffs.

The owner of the perpetual American put can exercise at any time. In particular, there is no expiration date after which the put can no longer be exercised. This makes every date like every other date; the time remaining to expiration is always the same (i.e., infinity). Because every date is like every other date, it is reasonable to expect that the optimal exercise policy depends only on the value of $S(t)$ and not on the time variable t . The owner of the put should exercise as soon as $S(t)$ falls far enough below K . In other words, it is reasonable to expect that the optimal exercise policy is of the form

Exercise the put as soon as $S(t)$ falls to the level L_* .

We have two questions to answer:

- (i) What is the value of L_* , and how do we know it corresponds to optimal exercise?
- (ii) What is the value of the put?

For the perpetual American put, we can base the answers to these questions on explicit computations.

21.1.1 Price under Arbitrary Exercise

Theorem 21.1: Laplace Transform For First Passage Time Of Drifted Brownian Motion

Let $\tilde{W}(t)$ be a Brownian motion under a probability measure $\tilde{\mathbb{P}}$, let μ be a real number, and let m be a positive number. Define $X(t) = \mu t + \tilde{W}(t)$, and set

$$\tau_m = \min\{t \geq 0; X(t) = m\},$$

If $X(t)$ never reaches the level m , then we interpret τ_m to be ∞ . Then

$$\tilde{\mathbb{E}}e^{-\lambda\tau_m} = e^{-m(-\mu + \sqrt{\mu^2 + 2\lambda})} \text{ for all } \lambda > 0, \quad (21.3)$$

where we interpret $e^{-\lambda\tau_m}$ to be zero if $\tau_m = \infty$.

Proof. Define $\sigma = -\mu + \sqrt{\mu^2 + 2\lambda}$ so that $\sigma > 0$ and

$$\begin{aligned} \sigma\mu + \frac{1}{2}\sigma^2 &= -\mu^2 + \mu\sqrt{\mu^2 + 2\lambda} + \frac{1}{2}\left(-\mu + \sqrt{\mu^2 + 2\lambda}\right)^2 \\ &= -\mu^2 + \mu\sqrt{\mu^2 + 2\lambda} + \frac{1}{2}\mu^2 - \mu\sqrt{\mu^2 + 2\lambda} + \frac{1}{2}\mu^2 + \lambda \\ &= \lambda. \end{aligned}$$

Then

$$e^{\sigma X(t) - \lambda t} = e^{\sigma\mu t + \sigma\tilde{W}(t) - \sigma\mu t - \frac{1}{2}\sigma^2 t} = e^{\sigma\tilde{W}(t) - \frac{1}{2}\sigma^2 t}.$$

This process is an exponential martingale (see [Theorem 6.11](#)) under $\tilde{\mathbb{P}}$ (its differential has a $d\tilde{W}(t)$ term and no dt term). According to [Theorem 5.18](#), the stopped martingale

$$M(t) = e^{\sigma\tilde{W}(t \wedge \tau_m) - \frac{1}{2}\sigma^2(t \wedge \tau_m)}$$

is also a martingale. Therefore, for each positive integer n ,

$$\begin{aligned}
 1 &= M(0) = \widetilde{\mathbb{E}}M(n) \\
 &= \widetilde{\mathbb{E}} \left[e^{\sigma X(n \wedge \tau_m) - \lambda(n \wedge \tau_m)} \right] \\
 &= \widetilde{\mathbb{E}} \left[e^{\sigma m - \lambda \tau_m} \mathbb{1}_{\{\tau_m \leq n\}} \right] + \widetilde{\mathbb{E}} \left[e^{\sigma X(n) - \lambda n} \mathbb{1}_{\{\tau_m > n\}} \right]. \tag{21.4}
 \end{aligned}$$

The nonnegative random variables $e^{\sigma m - \lambda \tau_m} \mathbb{1}_{\{\tau_m \leq n\}}$ increase with n , and their limit is $e^{\sigma m - \lambda \tau_m} \mathbb{1}_{\{\tau_m < \infty\}}$.

In other words,

$$0 \leq e^{\sigma m - \lambda \tau_m} \mathbb{1}_{\{\tau_m \leq 1\}} \leq e^{\sigma m - \lambda \tau_m} \mathbb{1}_{\{\tau_m \leq 2\}} \leq \dots \text{ almost surely,}$$

and

$$\lim_{n \rightarrow \infty} e^{\sigma m - \lambda \tau_m} \mathbb{1}_{\{\tau_m \leq n\}} = e^{\sigma m - \lambda \tau_m} \mathbb{1}_{\{\tau_m < \infty\}} \text{ almost surely.}$$

The Monotone Convergence Theorem, [Theorem 2.4](#) implies

$$\lim_{n \rightarrow \infty} \widetilde{\mathbb{E}} \left[e^{\sigma m - \lambda \tau_m} \mathbb{1}_{\{\tau_m \leq n\}} \right] = \widetilde{\mathbb{E}} \left[e^{\sigma m - \lambda \tau_m} \mathbb{1}_{\{\tau_m < \infty\}} \right]. \tag{21.5}$$

On the other hand, the random variable $e^{\sigma X(n) - \lambda n} \mathbb{1}_{\{\tau_m > n\}}$ satisfies

$$0 \leq e^{\sigma X(n) - \lambda n} \mathbb{1}_{\{\tau_m > n\}} \leq e^{\sigma m - \lambda n} \leq e^{\sigma m} \text{ almost surely}$$

because $X(n) \leq m$ for $n < \tau_m$ and σ is positive. Because λ is positive, we have

$$\lim_{n \rightarrow \infty} e^{\sigma X(n) - \lambda n} \mathbb{1}_{\{\tau_m > n\}} \leq \lim_{n \rightarrow \infty} e^{\sigma m - \lambda n} = 0.$$

According to the Dominated Convergence Theorem, [Theorem 2.5](#)

$$\lim_{n \rightarrow \infty} \widetilde{\mathbb{E}} \left[e^{\sigma X(n) - \lambda n} \mathbb{1}_{\{\tau_m > n\}} \right] = 0. \tag{21.6}$$

Taking the limit in Equation (21.4) and using Equations (21.5) and (21.6), we obtain

$$1 = \widetilde{\mathbb{E}} \left[e^{\sigma m - \lambda \tau_m} \mathbb{1}_{\{\tau_m < \infty\}} \right]$$

or, equivalently,

$$\widetilde{\mathbb{E}} \left[e^{-\lambda \tau_m} \mathbb{1}_{\{\tau_m < \infty\}} \right] = e^{-\sigma m} = e^{-m(-\mu + \sqrt{\mu^2 + 2\lambda})} \text{ for all } \lambda > 0. \quad (21.7)$$

This is Equation (21.3) when we interpret $e^{-\lambda \tau_m}$ to be zero if $\tau_m = \infty$. □

Tip 21.1: Limit Case for Lambda

We used the strict positivity of λ to derive (8.3.7), but now that we have it, we can take the limit as $\lambda \downarrow 0$. The random variables $e^{-\lambda \tau_m} \mathbb{1}_{\{\tau_m < \infty\}}$ are nonnegative and increase to $\mathbb{1}_{\{\tau_m < \infty\}}$ as $\lambda \downarrow 0$, and the Monotone Convergence Theorem, Theorem 2.4, allows us to conclude that

$$\widetilde{\mathbb{P}}\{\tau_m < \infty\} = \widetilde{\mathbb{E}} \mathbb{1}_{\{\tau_m < \infty\}} = \lim_{\lambda \downarrow 0} e^{-m(-\mu + \sqrt{\mu^2 + 2\lambda})} = e^{m\mu - m|\mu|}.$$

If $\mu \geq 0$, the drift in $X(t)$ is zero or upward, toward level m , and $\widetilde{\mathbb{P}}\{\tau_m < \infty\} = 1$; the level $X(t)$ is reached with probability one. On the other hand, if $\mu < 0$, the drift in $X(t)$ is downward, away from level m , and $\widetilde{\mathbb{P}}\{\tau_m < \infty\} = e^{-2m|\mu|} < 1$; there is a positive probability of never reaching m .

The solution to Equation (21.1) is

$$S(t) = S(0) \exp\left\{\sigma \widetilde{W}(t) + \left(r - \frac{1}{2}\sigma^2\right)t\right\}. \quad (21.8)$$

Suppose the owner of the perpetual American put sets a positive level $L < K$ and resolves to exercise the put the first time the stock price falls to L . If the initial stock price is at or below L , she exercises immediately (at time zero). The value of the put in this case is

$v_L(S(0)) = K - S(0)$. If the initial stock price is above L , she exercises at the stopping time

$$\tau_L = \min\{t \geq 0; S(t) = L\}, \quad (21.9)$$

where τ_L is set equal to ∞ if the stock price never reaches the level L . At the time of exercise, the put pays $K - S(\tau_L) = K - L$. Discounting this back to time zero and taking the risk-neutral expected value, we compute the value of the put under this exercise strategy to be

$$v_L(S(0)) = (K - L)\widetilde{\mathbb{E}}e^{-r\tau_L} \text{ for all } S(0) \geq L. \quad (21.10)$$

On those paths where $\tau_L = \infty$, we interpret $e^{-r\tau_L}$ to be zero. (Recall our assumption at the beginning of this section that r is strictly positive.) Although not explicitly indicated by the notation, the distribution of τ_L depends on the initial stock price $S(0)$, so the right-hand side (21.10) is a function of $S(0)$.

Lemma 21.1: Value Function Formula

The function $v_L(x)$ is given by the formula

$$v_L(x) = \begin{cases} K - x, & 0 \leq x \leq L, \\ (K - L) \left(\frac{x}{L}\right)^{-\frac{2r}{\sigma^2}}, & x \geq L. \end{cases} \quad (21.11)$$

Proof. We only need to establish the second line of Equation (21.11). If $x = L$, then $\tau_L = 0$ and Equation (21.10) implies $v_L(x) = K - L$.

We consider the case $S(0) = x > L$. The stopping time τ_L is the first time $S(t)$ reaches the level L . But $S(t) = L$ if and only if

$$S(t) = x \exp \left\{ \sigma \widetilde{W}(t) + \left(r - \frac{1}{2} \sigma^2 \right) t \right\}$$

implies

$$-\widetilde{W}(t) - \frac{1}{\sigma} \left(r - \frac{1}{2} \sigma^2 \right) t = \frac{1}{\sigma} \log \frac{x}{L}.$$

We now apply [Theorem 21.1](#) with $X(t)$ in that theorem replaced by $-\tilde{W}(t) - \frac{1}{\sigma} \left(r - \frac{1}{2}\sigma^2\right) t$ (the processes $\tilde{W}(t)$ and $-\tilde{W}(t)$ are both Brownian motions under $\tilde{\mathbb{P}}$), with λ replaced by r , with μ replaced by $-\frac{1}{\sigma} \left(r - \frac{1}{2}\sigma^2\right)$, and with m replaced by $\frac{1}{\sigma} \log \frac{x}{L}$, which is positive. With these replacements, τ_m in [Theorem 21.1](#) is τ_L and

$$\begin{aligned}\mu^2 + 2\lambda &= \frac{1}{\sigma^2} \left(r^2 - r\sigma^2 + \frac{1}{4}\sigma^4 \right) + 2r \\ &= \frac{1}{\sigma^2} \left(r^2 + r\sigma^2 + \frac{1}{4}\sigma^4 \right) \\ &= \frac{1}{\sigma^2} \left(r + \frac{1}{2}\sigma^2 \right)^2.\end{aligned}$$

Therefore,

$$-\mu + \sqrt{\mu^2 + 2\lambda} = \frac{1}{\sigma} \left(r - \frac{1}{2}\sigma^2 \right) + \frac{1}{\sigma} \left(r + \frac{1}{2}\sigma^2 \right) = \frac{2r}{\sigma}.$$

[Equation \(21.3\)](#) implies

$$\tilde{\mathbb{E}}[e^{-r\tau_L}] = \exp \left\{ -\frac{1}{\sigma} \log \frac{x}{L} \cdot \frac{2r}{\sigma} \right\} = \left(\frac{x}{L} \right)^{-\frac{2r}{\sigma^2}}.$$

The second line in [Equation \(21.11\)](#) follows. □

21.1.2 Price under Optimal Exercise

[Figure 6](#) shows the function $v_L(x)$ for three different values of L . The function $v_{L_1}(x)$ in that figure actually lies below the intrinsic value $K - x$ for x between L_1 and L_2 . If the initial stock price is between L_1 and L_2 , then the strategy of exercising the first time the stock price falls to L_1 is obviously a poor one; it would be better to exercise at time zero and receive the intrinsic value. The function $v_{L_2}(x)$ agrees with the intrinsic value for $0 \leq x \leq L_2$ and follows the indicated curve for $x \geq L_2$. The function $v_{L_*}(x)$ agrees with the intrinsic value for $0 \leq x \leq L_*$ and follows the indicated curve for $x \geq L_*$. For $x \geq L_*$, the function $v_{L_*}(x)$ is strictly larger than the function $v_{L_2}(x)$, and hence the strategy of exercising the first time the stock price falls to L_* is better than exercising the first time the stock price falls to L_2 .

PerpetualPut.py

```

1  # Model Parameters
2  K, alpha = 10.0, 0.82
3  L_star = (alpha / (1 + alpha)) * K # Optimal exercise boundary
4  x = np.linspace(0.1, 15.0, 500)    # State space grid
5
6  # Valuation
7  v = np.where(x > L_star,
8  (K - L_star) * (x / L_star) ** -alpha,
9  K - x)

```

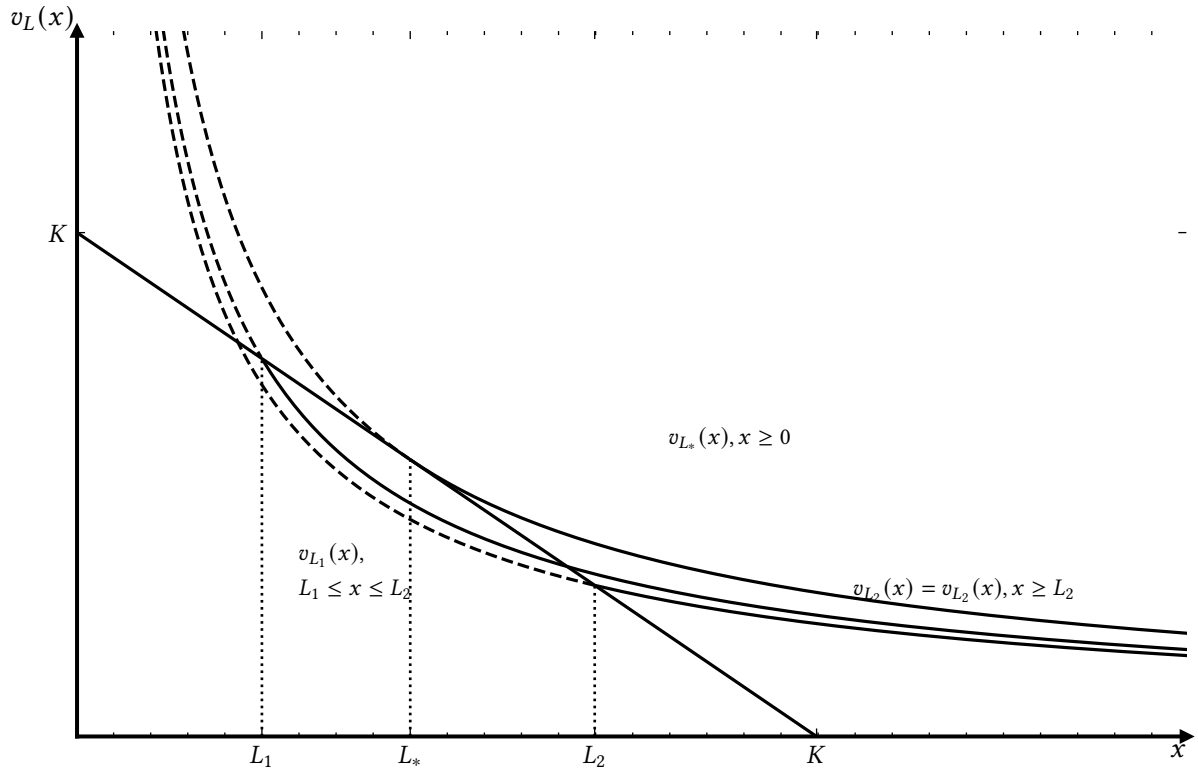


Figure 6: $(K - L)\left(\frac{x}{L}\right)^{-\frac{2r}{\sigma^2}}$ for three values of L .

As Figure 6 suggests, for any value of L smaller than L_* , the function $v_L(x)$ agrees with the intrinsic value for $0 \leq x \leq L$, lies below the intrinsic value immediately to the right of L , and lies below $v_{L_*}(x)$ everywhere to the right of L . For any value of L larger than L_* , the function $v_L(x)$ agrees with the intrinsic value for $0 \leq x \leq L$ and lies below $v_{L_*}(x)$ for all $x \geq L_*$. Thus, among those exercise policies of the form

“Exercise the put as soon as $S(t)$ falls to the level L ,”

the best one is obtained by choosing $L = L_*$. We expect therefore that $v_{L_*}(x)$ is the price of the put $v_*(x)$ of [Definition 21.1](#). We prove this below.

We must first determine the value of L_* . We note that

$$v_L(x) = (K - L)L^{\frac{2r}{\sigma^2}}x^{-\frac{2r}{\sigma^2}} \quad \text{for all } x \geq L. \quad (21.12)$$

From [Figure 6](#), we know that L_* is the value of L that maximizes this quantity when we hold x fixed. We thus define

$$g(L) = (K - L)L^{\frac{2r}{\sigma^2}} \quad (21.13)$$

and seek the value of L that maximizes this function over $L \geq 0$. Because $\frac{2r}{\sigma^2}$ is strictly positive, we have $g(0) = 0$ and $\lim_{L \rightarrow \infty} g(L) = -\infty$. Moreover,

$$g'(L) = -L^{\frac{2r}{\sigma^2}} + \frac{2r}{\sigma^2}(K - L)L^{\frac{2r}{\sigma^2}-1} = -\frac{2r + \sigma^2}{\sigma^2}L^{\frac{2r}{\sigma^2}} + \frac{2r}{\sigma^2}KL^{\frac{2r}{\sigma^2}-1}. \quad (21.14)$$

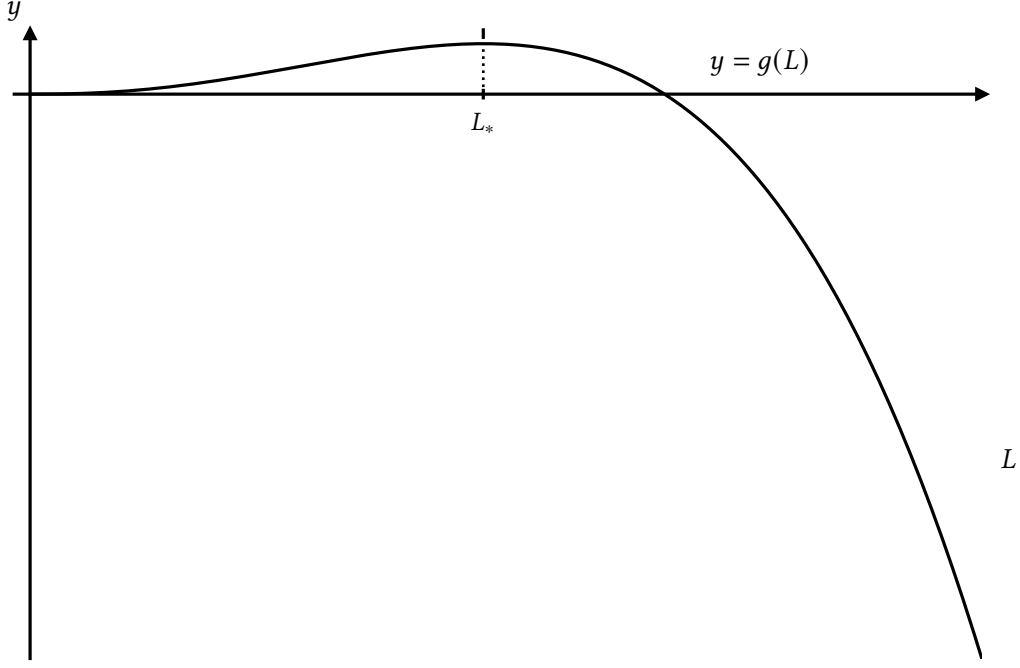
Setting this equal to zero, we solve for

$$L_* = \frac{2r}{2r + \sigma^2}K. \quad (21.15)$$

This is a number between 0 and K . Furthermore,

$$g(L_*) = \frac{\sigma^2}{2r + \sigma^2} \left(\frac{2r}{2r + \sigma^2} \right)^{\frac{2r}{\sigma^2}} K^{\frac{2r + \sigma^2}{\sigma^2}} \quad (21.16)$$

is strictly positive. Therefore, the graph of $y = g(L)$ must be as shown in [Figure 7](#) and L_* given by [Equation \(21.15\)](#) is the point where $g(L)$ attains its maximum.

Figure 7: Graph of $g(L)$.

21.1.3 Analytical Characterisation of the Put Price

We have

$$v_{L_*}(x) = \begin{cases} K - x, & 0 \leq x \leq L_*, \\ (K - L_*) \left(\frac{x}{L_*} \right)^{-\frac{2r}{\sigma^2}}, & x \geq L_*, \end{cases} \quad (21.17)$$

so that

$$v'_{L_*}(x) = \begin{cases} -1, & 0 \leq x \leq L_*, \\ -(K - L_*) \frac{2r}{\sigma^2 x} \left(\frac{x}{L_*} \right)^{-\frac{2r}{\sigma^2}}, & x \geq L_*. \end{cases} \quad (21.18)$$

If we evaluate the second line in Equation (21.18) at $x = L_*$, we get the right-hand derivative

$$v'_{L_*}(L_*) = -\frac{2r}{\sigma^2 L_*} (K - L_*) = -\frac{2rK}{\sigma^2 L_*} + \frac{2r}{\sigma^2} = -\frac{2r}{\sigma^2} \cdot \frac{2r + \sigma^2}{2r} + \frac{2r}{\sigma^2} = -1,$$

which agrees with the left-hand derivative $v'_{L_*}(L_*-) = -1$ provided by the first line in Equation (21.18). The derivative of $v_{L_*}(x)$ is continuous at $x = L_*$. This is known as *smooth pasting*. The two parts of the definition of $v_{L_*}(x)$ fit together at $x = L_*$ so that both $v_{L_*}(x)$

and $v'_{L_*}(x)$ are continuous. This is because the graph of the function $y = (K - L_*)\left(\frac{x}{L_*}\right)^{-\frac{2r}{\sigma^2}}$ is tangent to the line $y = K - x$ at $x = L_*$, as one can see from Figure 6. In fact, we could have used the smooth pasting condition to solve for L_* .

The second derivative of $v(x)$ has a jump at $x = L_*$, and hence is undefined at this point. Indeed,

$$v''_{L_*}(x) = \begin{cases} 0, & 0 \leq x < L_*, \\ (K - L_*) \frac{2r(2r + \sigma^2)}{\sigma^4 x^2} \left(\frac{x}{L_*}\right)^{-\frac{2r}{\sigma^2}}, & x > L_*. \end{cases} \quad (21.19)$$

The left-hand and right-hand second derivatives at $x = L_*$ are $v''(L_*-) = 0$ and $v''(L_*+) = (K - L_*) \frac{2r(2r + \sigma^2)}{\sigma^4 L_*^2} > 0$.

For $x > L_*$, we can verify by direct computation that

$$\begin{aligned} & rv_{L_*}(x) - rxv'_{L_*}(x) - \frac{1}{2}\sigma^2 x^2 v''_{L_*}(x) \\ &= (K - L_*) \left(r + \frac{2r^2}{\sigma^2} - \frac{r(2r + \sigma^2)}{\sigma^2} \right) \left(\frac{x}{L_*} \right)^{-\frac{2r}{\sigma^2}} = 0. \end{aligned} \quad (21.20)$$

On the other hand, for $0 \leq x < L_*$,

$$rv_{L_*}(x) - rxv'_{L_*}(x) - \frac{1}{2}\sigma^2 x^2 v''_{L_*}(x) = r(K - x) + rx = rK. \quad (21.21)$$

In particular, we see that $v_{L_*}(x)$ satisfies the so-called *linear complementarity conditions*

$$v(x) \geq (K - x)^+ \text{ for all } x \geq 0, \quad (21.22)$$

$$rv(x) - rxv'(x) - \frac{1}{2}\sigma^2 x^2 v''(x) \geq 0 \text{ for all } x \geq 0, \text{ and} \quad (21.23)$$

$$\text{for each } x \geq 0, \text{ equality holds in either Equation (21.22) or Equation (21.23).} \quad (21.24)$$

The point L_* is slightly problematical in Equation (21.23) since $v''_{L_*}(L_*)$ is undefined. However, if we replace $v''_{L_*}(L_*)$ in Equation (21.23) by either $v''_{L_*}(L_*-)$ or $v''_{L_*}(L_*+)$, the inequality holds.

The linear complementarity conditions [Equations \(21.22\) to \(21.24\)](#) determine the function $v_{L_*}(x)$. More precisely, the function $v_{L_*}(x)$ given by [Equation \(21.17\)](#) is the only bounded continuous function having a continuous derivative that satisfies these conditions.

21.1.4 Probabilistic Characterisation of the Put Price

Theorem 21.2: Stock Price Supermartingale

Let $S(t)$ be the stock price given by [Equation \(21.1\)](#) and let τ_{L_*} be given by [Equation \(21.9\)](#) with $L = L_*$. Then $e^{-rt}v_{L_*}(S(t))$ is a supermartingale under $\tilde{\mathbb{P}}$, and the stopped process $e^{-r(t \wedge \tau_{L_*})}v_{L_*}(S(t \wedge \tau_{L_*}))$ is a martingale.

Proof. Fortunately, the Itô formula applies to functions whose second derivatives have jumps, provided the first derivative is continuous. We may thus compute

$$\begin{aligned} d[e^{-rt}v_{L_*}(S(t))] &= e^{-rt} \left[-rv_{L_*}(S(t)) dt + v'_{L_*}(S(t)) dS(t) + \frac{1}{2}v''_{L_*}(S(t)) dS(t) dS(t) \right] \\ &= e^{-rt} \left[-rv_{L_*}(S(t)) + rS(t)v'_{L_*}(S(t)) + \frac{1}{2}\sigma^2 S^2(t)v''_{L_*}(S(t)) \right] dt \\ &\quad + e^{-rt}\sigma S(t)v'_{L_*}(S(t)) d\tilde{W}(t). \end{aligned}$$

Because of [Equations \(21.20\) to \(21.21\)](#), the dt term in this expression is either 0 or $-rK$, depending on whether $S(t) > L_*$ or $S(t) < L_*$. If $S(t) = L_*$, $v''_{L_*}(S(t))$ is undefined, but the probability $S(t) = L_*$ is zero so this does not matter. We thus have

$$d[e^{-rt}v_{L_*}(S(t))] = -e^{-rt}rK\mathbf{I}_{\{S(t) < L_*\}} dt + e^{-rt}\sigma S(t)v'_{L_*}(S(t)) d\tilde{W}(t). \quad (21.25)$$

Because the dt term in [Equation \(21.25\)](#) is less than or equal to zero, $e^{-rt}v_{L_*}(S(t))$ is a supermartingale; when $S(t) < L_*$ it has a downward tendency. If the initial stock price is above L_* , then prior to the time τ_{L_*} when the stock price first reaches L_* , the dt term in [Equation \(21.25\)](#) is zero and hence $e^{-r(t \wedge \tau_{L_*})}v(S(t \wedge \tau_{L_*}))$ is a martingale. Indeed, integration

of Equation (21.25) yields

$$e^{-r(t \wedge \tau_{L_*})} v_{L_*}(S(t \wedge \tau_{L_*})) = v_{L_*}(0) + \int_0^{t \wedge \tau_{L_*}} e^{-ru} \sigma S(u) v'_{L_*}(S(u)) d\tilde{W}(u).$$

Itô integrals are martingales, and hence the Itô integral above stopped at the stopping time τ_{L_*} is a martingale. $\square$

Corollary 21.1: Perpetual American Put Price

Recall that $\mathcal{T}$ is the set of all stopping times, not just those of the form (8.3.9). We have

$$v_{L_*}(x) = \max_{\tau \in \mathcal{T}} \tilde{\mathbb{E}}[e^{-r\tau}(K - S(\tau))],$$

where $x = S(0)$ is the initial stock price. In other words, $v_{L_*}(x)$ is the perpetual American put price of Definition 21.1.

Proof. Because $e^{-rt} v_{L_*}(S(t))$ is a supermartingale under $\tilde{\mathbb{P}}$, we have from Theorem 5.18 that, for every stopping time $\tau \in \mathcal{T}$,

$$v_{L_*}(x) = v_{L_*}(S(0)) \geq \tilde{\mathbb{E}}[e^{-r(t \wedge \tau)} v_{L_*}(S(t \wedge \tau))]. \quad (21.26)$$

Because $v_{L_*}(S(t \wedge \tau))$ is bounded, we may let $t \rightarrow \infty$ in Equation (21.26), using the Dominated Convergence Theorem, Theorem 2.5, to conclude that

$$v_{L_*}(x) \geq \tilde{\mathbb{E}}[e^{-r\tau} v_{L_*}(S(\tau))] \geq \tilde{\mathbb{E}}[e^{-r\tau}(K - S(\tau))],$$

where we have gotten the last inequality from Equation (21.22). Because this inequality holds for every $\tau \in \mathcal{T}$, we have

$$v_{L_*}(x) \geq \max_{\tau \in \mathcal{T}} \tilde{\mathbb{E}}[e^{-r\tau}(K - S(\tau))].$$

On the other hand, if we replace τ by τ_{L_*} , we obtain equality in Equation (21.26) because $e^{-r(t \wedge \tau_{L_*})} v_{L_*}(S(t \wedge \tau_{L_*}))$ is a martingale under $\tilde{\mathbb{P}}$. Letting $t \rightarrow \infty$ and using the Dominated

Convergence Theorem, [Theorem 2.5](#), we obtain

$$v_{L_*}(x) = \widetilde{\mathbb{E}} \left[e^{-r\tau_{L_*}} v_{L_*}(S(\tau_{L_*})) \right].$$

Since

$$e^{-r\tau_{L_*}} v_{L_*}(S(\tau_{L_*})) = e^{-r\tau_{L_*}} v_{L_*}(L_*) = e^{-r\tau_{L_*}} (K - L_*) = e^{-r\tau_{L_*}} (K - S(\tau_{L_*}))$$

if $\tau_{L_*} < \infty$ (and is interpreted to be zero if $\tau_{L_*} = \infty$), we see that

$$v_{L_*}(x) = \widetilde{\mathbb{E}} \left[e^{-r\tau_{L_*}} (K - S(\tau_{L_*})) \right]. \quad (21.27)$$

It follows that

$$v_{L_*}(x) \leq \max_{\tau \in \mathcal{T}} \widetilde{\mathbb{E}} \left[e^{-r\tau} (K - S(\tau)) \right].$$

□

Discounted European option prices are martingales under the risk-neutral probability measure. Discounted American option prices are martingales up to the time they should be exercised. If they are not exercised when they should be, they tend downward. Since a martingale is a special case of a supermartingale, and processes that tend downward are supermartingales, discounted American option prices are supermartingales. An agent who is short an American option can hedge that short position in the usual way during the time the discounted option price is a martingale. If the option is not exercised when it should be, then the agent can continue the hedge and take money off the table. The following corollary illustrates this for the perpetual American put of this section.

Corollary 21.2: Perpetual American Put Hedging

Consider an agent with initial capital $X(0) = v_{L_*}(S(0))$, the initial perpetual American put price. Suppose this agent uses the portfolio process $\Delta(t) = v'_{L_*}(S(t))$ and consumes cash at rate $C(t) = rK\mathbb{I}_{\{S(t) < L_*\}}$ (i.e., consumes cash at rate rK whenever $S(t) < L_*$). Then the value $X(t)$ of the agent's portfolio agrees with the option price $v_{L_*}(S(t))$ for all times t until the option is exercised. In particular, $X(t) \geq (K - S(t))^+$ for all t until the option is exercised, so the agent can pay off a short option position regardless of when the option is exercised.

Proof. The differential of the agent's portfolio value process is

$$dX(t) = \Delta(t) dS(t) + r(X(t) - \Delta(t)S(t)) dt - C(t) dt,$$

so the differential of the discounted portfolio value process is

$$\begin{aligned} d(e^{-rt}X(t)) &= e^{-rt}(-rX(t) dt + dX(t)) \\ &= e^{-rt}(\Delta(t) dS(t) - r\Delta(t)S(t) dt - C(t) dt) \\ &= e^{-rt}(\Delta(t)\sigma S(t) d\tilde{W}(t) - C(t) dt). \end{aligned} \tag{21.28}$$

Substituting $\Delta(t) = v'_{L_*}(S(t))$ and $C(t) = rK\mathbb{I}_{\{S(t) < L_*\}}$ into Equation (21.28) and comparing it to Equation (21.25), we see that $d(e^{-rt}X(t)) = d[e^{-rt}v_{L_*}(S(t))]$. Integrating both sides of this equation and using the initial equality $X(0) = v_{L_*}(S(0))$, we obtain $X(t) = v_{L_*}(S(t))$ for all t prior to exercise. $\square$

Tip 21.2: Hedging Mechanics

During any period in which $S(t) < L_*$, the agent in Corollary 21.2 has stock position $\Delta(t) = v'_{L_*}(S(t)) = -1$ (i.e., is short one share of stock) and has a total portfolio value $X(t) = v_{L_*}(S(t)) = K - S(t)$. Therefore, the agent has K invested in the money market. If the owner of the put exercises, the agent in Corollary 21.2 receives a share of stock, which covers his short position, and pays out K from his money market





account. If the owner of the put does not exercise, the agent holds his position and consumes the interest from the money market investment (i.e., consumes cash at rate rK per unit time).

The argument in [Corollary 21.2](#) applies generally. In a complete market, whenever some discounted price process is a supermartingale, it is possible to construct a hedging portfolio whose value tracks the price process. This portfolio may sometimes consume. In the case of the perpetual American put, the supermartingale property for the discounted put price follows from [Equation \(21.23\)](#). If, in addition, the price process dominates some intrinsic value (see [Equation \(21.22\)](#) for the perpetual American put), then a short position in the American option with that intrinsic value can be hedged. There are always two conditions on the price of any American option, corresponding to [Equation \(21.22\)](#) and [Equation \(21.23\)](#). These conditions guarantee that the price is sufficient to satisfy the seller of the put.

However, [Equations \(21.22\) to \(21.23\)](#) alone are not enough to determine the price of the perpetual American put. There can be functions that satisfy these conditions but are strictly greater than the price $v_{L^*}(x)$ we constructed in [Equation \(21.17\)](#). There must be some additional condition that guarantees that the price is satisfactory for the purchaser of the put. One version of this condition for the perpetual American put is [Equation \(21.24\)](#), guaranteeing that there exists an exercise strategy that permits the owner of the put to capture the full value of the put. It says that if we divide the half-line $[0, \infty)$ into two sets, the *stopping set*

$$S = \{x \geq 0; v_{L^*}(x) = (K - x)^+\} \quad (21.29)$$

and the *continuation set*

$$C = \{x \geq 0; v_{L^*}(x) > (K - x)^+\}, \quad (21.30)$$

then equality holds in [Equation \(21.23\)](#) for $x \in C$. If the initial stock price is in S , then the owner of the put can get full value by exercising it immediately. On the other hand, if the

initial stock price is in C , then the put is more valuable than its intrinsic value, and the owner of the put can capture this extra value by waiting until the stock price enters S to exercise, if it ever does enter S . The time of entry into the set S is in fact τ_{L^*} in [Theorem 21.2](#). We saw in [Equation \(21.27\)](#) that

$$v(S(0)) = \tilde{\mathbb{E}} \left[e^{-r\tau^*} v(S(\tau^*)) \right] = \tilde{\mathbb{E}} \left[e^{-r\tau^*} (K - S(\tau^*)) \right].$$

In conclusion, the three linear complementarity conditions have counterparts that can be stated probabilistically rather than analytically (i.e., without writing conditions on the derivatives of $v(x)$). Let $V(t) = e^{-rt}v(S(t))$ be the value of the perpetual American put. The stochastic process $V(t)$ satisfies the following three conditions:

- (i) $V(t) \geq (K - S(t))^+$ for all $t \geq 0$,
- (ii) $e^{-rt}V(t)$ is a supermartingale under $\tilde{\mathbb{P}}$, and
- (iii) there exists a stopping time τ_* such that

$$V(0) = \tilde{\mathbb{E}} \left[e^{-r\tau_*} (K - S(\tau_*))^+ \right].$$

These three conditions determine the value of $V(0)$.

21.2 Finite-Expiration American Put

21.2.1 Analytical Characterisation of the Put Price

21.2.2 Probabilistic Characterisation of the Put Price

21.3 American Call

We treat the American call, first on the usual geometric Brownian motion asset, [Equation \(21.1\)](#) and then on a variation of this asset that pays dividends at discrete dates. In the first case, we see that the American call price is the same as the European call price. In the second case, we provide a recursion formula for computing the American call price.

21.3.1 Underlying Asset Pays No Dividends

We begin with a case slightly more general than a call option. Consider a stock whose price process $S(t)$ is given by

$$dS(t) = rS(t)dt + \sigma S(t)d\tilde{W}(t), \quad (21.31)$$

where the interest rate r and the volatility σ are strictly positive and $\tilde{W}(t)$ is a Brownian motion under the risk-neutral probability measure $\tilde{\mathbb{P}}$.

Lemma 21.2: Discounted Intrinsic Value Submartingale

Let $h(x)$ be a nonnegative, convex function of $x \geq 0$ satisfying $h(0) = 0$. Then the discounted intrinsic value $e^{-rt}h(S(t))$ of the American derivative security that pays $h(S(t))$ upon exercise is a submartingale.

Proof. Because $h(x)$ is convex, for $0 \leq \lambda \leq 1$ and $0 \leq x_1 \leq x_2$, we have

$$h((1 - \lambda)x_1 + \lambda x_2) \leq (1 - \lambda)h(x_1) + \lambda h(x_2). \quad (21.32)$$

See Figure 8.5.1 for the case of a call payoff, $h(x) = (x - K)^+$. Taking $x_1 = 0$, $x_2 = x$, and using the fact that $h(0) = 0$, we obtain from Equation (21.32) that

$$h(\lambda x) \leq \lambda h(x) \quad \text{for all } x \geq 0, 0 \leq \lambda \leq 1. \quad (21.33)$$

For $0 \leq u \leq t \leq T$, we have $0 \leq e^{-r(t-u)} \leq 1$, and Equation (21.33) implies

$$\tilde{\mathbb{E}} \left[e^{-r(t-u)} h(S(t)) \middle| \mathcal{F}(u) \right] \geq \tilde{\mathbb{E}} \left[h \left(e^{-r(t-u)} S(t) \right) \middle| \mathcal{F}(u) \right]. \quad (21.34)$$

The conditional Jensen's inequality, Lemma 2.2 implies

$$\tilde{\mathbb{E}} \left[h \left(e^{-r(t-u)} S(t) \right) \middle| \mathcal{F}(u) \right] \geq h \left(\tilde{\mathbb{E}} \left[e^{-r(t-u)} S(t) \middle| \mathcal{F}(u) \right] \right) = h \left(e^{ru} \tilde{\mathbb{E}} \left[e^{-rt} S(t) \middle| \mathcal{F}(u) \right] \right). \quad (21.35)$$

Because $e^{-rt}S(t)$ is a martingale under $\widetilde{\mathbb{P}}$, we have

$$h\left(e^{ru}\widetilde{\mathbb{E}}\left[e^{-rt}S(t)\middle|\mathcal{F}(u)\right]\right) = h(e^{ru}e^{-ru}S(u)) = h(S(u)). \quad (21.36)$$

Putting Equations (21.34) to (21.36) together, we conclude that

$$\widetilde{\mathbb{E}}\left[e^{-r(t-u)}h(S(t))\middle|\mathcal{F}(u)\right] \geq h(S(u)) \quad (21.37)$$

or, equivalently,

$$\widetilde{\mathbb{E}}\left[e^{-rt}h(S(t))\middle|\mathcal{F}(u)\right] \geq e^{-ru}h(S(u)). \quad (21.38)$$

This is the submartingale property for $e^{-rt}h(S(t))$. $\square$

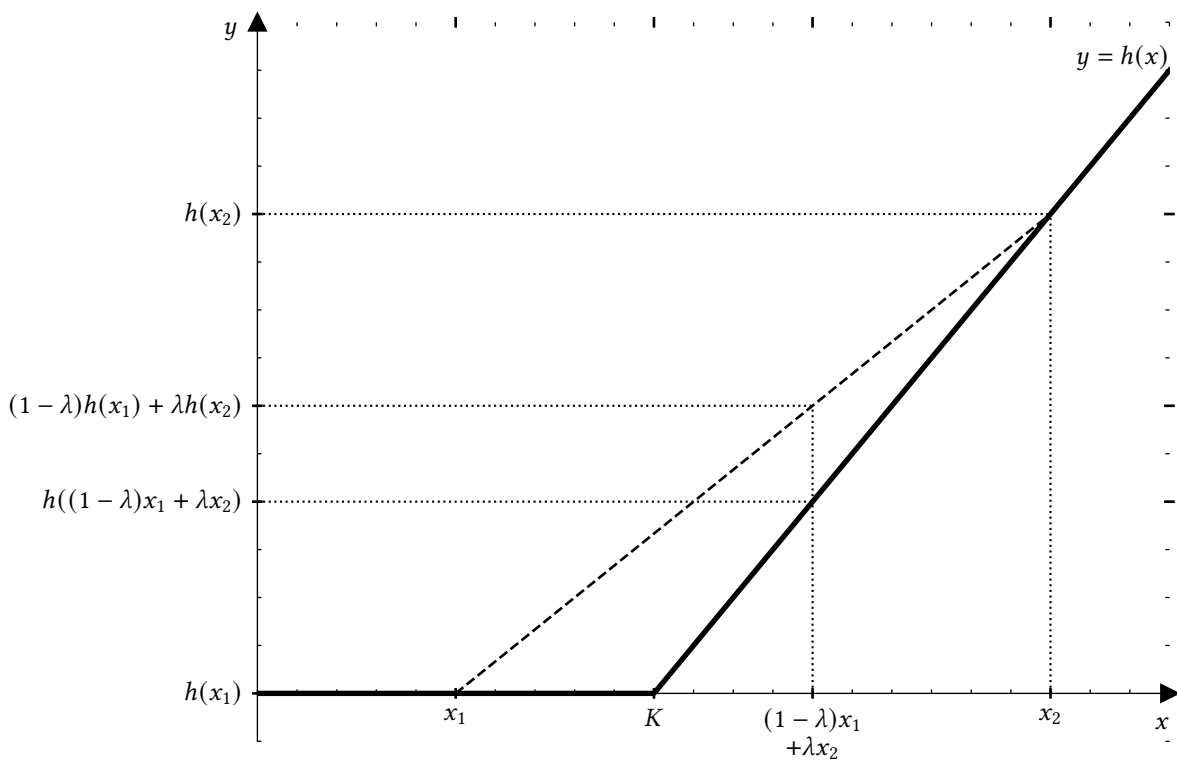


Figure 8: The convex function $h(x) = (x - K)^+$.

Theorem 21.3: Price of American Derivative

Let $h(x)$ be a nonnegative, convex function of $x \geq 0$ satisfying $h(0) = 0$. Then the price of the American derivative security expiring at time T and having intrinsic value $h(S(t))$, $0 \leq t \leq T$, is the same as the price of the European derivative security paying $h(S(T))$ at expiration T .

Proof. Replacing t by T in Equation (21.37), we obtain

$$\tilde{\mathbb{E}} \left[e^{-r(T-u)} h(S(T)) \mid \mathcal{F}(u) \right] \geq h(S(u)), \quad 0 \leq u \leq T.$$

In other words, the European derivative security price always dominates the intrinsic value of the American derivative security. This shows that the option to exercise early is worthless, and the price of the American derivative security agrees with the price of the European security. $\square$

Corollary 21.3: American Call Price

The price of an American call on an asset not paying a dividend is the same as the price of the European call on the same asset with the same expiration.

Proof. Take $h(x) = (x - K)^+$ in Theorem 21.3. $\square$

The idea behind Corollary 21.3 is that the discounted process $e^{-rt}(S(t) - K)^+$ is a submartingale under $\tilde{\mathbb{P}}$ and hence tends to rise. Therefore, it is optimal to wait until expiration before deciding whether to exercise.

There are two factors that contribute to the submartingale property for $e^{-rt}(S(t) - K)^+$. One is the discounting of the strike. In fact, $e^{-rt}(S(t) - K)$ (without the $^+$) is a submartingale because $e^{-rt}S(t)$ is a martingale under the risk-neutral measure $\tilde{\mathbb{P}}$ and $-e^{-rt}K$ increases as t increases (throughout this chapter, we assume a strictly positive interest rate r). When we reinstate the $^+$, we are taking a convex function of a submartingale and, because of Jensen's inequality, this reinforces the upward trend.

The previous argument does not apply to the American put, whose discounted intrinsic value $e^{-rt}(K - S(t))$ (without the $^+$) is a supermartingale ($e^{-rt}K$ falls and $-e^{-rt}S(t)$ is a

martingale). Jensen's inequality creates an upward trend that competes with this supermartingale property, and the analysis becomes complicated.

If the underlying asset pays a dividend, the case considered in the next subsection, the argument above no longer applies to the American call. In this case, $e^{-rt}S(t)$ is a supermartingale and tends to fall because of the dividend outflow.

21.3.2 Underlying Asset Pays Dividends

In this subsection, we consider an American call on an asset whose price process is a geometric Brownian motion governed by (8.5.1) between dividend payment dates.

We assume there are times $0 < t_1 < t_2 < \cdots < t_n < T$, and at each time t_j the dividend paid is $a_j S(t_j-)$, where $S(t_j-)$ denotes the asset price just prior to the dividend payment. The asset price $S(t_j)$ after the dividend payment is the asset price before the dividend payment less the dividend payment:

$$S(t_j) = S(t_j-) - a_j S(t_j-) = (1 - a_j)S(t_j-). \quad (21.39)$$

We assume that each $a_j, j = 1, \dots, n$, is a number between 0 and 1. We set $t_0 = 0$, but this is not a dividend payment date. We also assume that T is not a dividend payment date, although it is not difficult to modify the analysis given below to handle the case when T is a dividend payment date.

We shall see that it is not optimal to exercise an American call on this asset except possibly immediately before a dividend payment. The price of the call will be seen to satisfy the Black-Scholes-Merton partial differential equation between dividend payment dates. At dividend payment dates, the price of the call is the maximum of the call's intrinsic value and the price of the call after the dividend is paid and the stock price is reduced by the amount of the payment. These observations lead to a recursive algorithm for determining the price, and that is developed in this subsection.

For $t_j \leq t < t_{j+1}$, we have

$$S(t) = S(t_j) \exp \left\{ \sigma (\tilde{W}(t) - \tilde{W}(t_j)) + \left(r - \frac{1}{2} \sigma^2 \right) (t - t_j) \right\}, \quad (21.40)$$

which implies

$$S(t_{j+1}-) = S(t_j) \exp \left\{ \sigma (\tilde{W}(t_{j+1}) - \tilde{W}(t_j)) + \left(r - \frac{1}{2} \sigma^2 \right) (t_{j+1} - t_j) \right\} \quad (21.41)$$

and

$$S(t_{j+1}) = (1 - a_{j+1}) S(t_j) \exp \left\{ \sigma (\tilde{W}(t_{j+1}) - \tilde{W}(t_j)) + \left(r - \frac{1}{2} \sigma^2 \right) (t_{j+1} - t_j) \right\}. \quad (21.42)$$

We also have

$$S(T) = S(t_n) \exp \left\{ \sigma (\tilde{W}(T) - \tilde{W}(t_n)) + \left(r - \frac{1}{2} \sigma^2 \right) (T - t_n) \right\}. \quad (21.43)$$

We consider an American call expiring at time T with strike price K . For $t_n \leq t \leq T$, the discounted asset price $e^{-rt} S(t)$ is a martingale under $\tilde{\mathbb{P}}$, and [Lemma 21.2](#) can be invoked to show that $e^{-rt} (S(t) - K)^+$ is a submartingale. Therefore,

$$\tilde{\mathbb{E}} \left[e^{-rT} (S(T) - K)^+ \mid \mathcal{F}(t) \right] \geq e^{-rt} (S(t) - K)^+, \quad t_n \leq t \leq T. \quad (21.44)$$

This shows that, for all $t \in [t_n, T]$, the price of the European call at time t ,

$$c_n(t, S(t)) = \tilde{\mathbb{E}} \left[e^{-r(T-t)} (S(T) - K)^+ \mid \mathcal{F}(t) \right], \quad (21.45)$$

is greater than the intrinsic value of the American call, $(S(t) - K)^+$. Consequently, the early exercise feature of the American call is worthless, and the prices at time t of the European and American calls agree for $t_n \leq t \leq T$. This price is given by the Black-Scholes-Merton formula

$$c_n(t, x) = x N(d_+(T-t, x)) - K e^{-r(T-t)} N(d_-(T-t, x)), \quad (21.46)$$

where

$$d_{\pm}(\tau, x) = \frac{1}{\sigma \sqrt{\tau}} \left[\log \frac{x}{K} + \left(r \pm \frac{1}{2} \sigma^2 \right) \tau \right].$$

Although one cannot simply substitute $x = 0$ into (8.5.14), we have $c(t, 0) = 0$; see equation (4.5.17) and Exercise 4.9. Formula (8.5.14) can be determined by computing the conditional expectation in (8.5.13) under the condition $S(t) = x$. In the case $t = t_n$, using (8.5.12), this leads to

$$c_n(t_n, x) = \widetilde{\mathbb{E}} \left[e^{-r(T-t_n)} \left(x \exp \left\{ \sigma(\widetilde{W}(T) - \widetilde{W}(t_n)) + \left(r - \frac{1}{2}\sigma^2 \right) (T - t_n) \right\} - K \right)^+ \right]. \quad (21.47)$$

The function $c_n(t, x)$ also satisfies the Black-Scholes-Merton differential equation

$$\frac{\partial}{\partial t} c_n(t, x) + rx \frac{\partial}{\partial x} c_n(t, x) + \frac{1}{2} \sigma^2 x^2 \frac{\partial^2}{\partial x^2} c_n(t, x) = r c_n(t, x), \quad t_n \leq t < T, \quad x \geq 0, \quad (21.48)$$

and the terminal condition

$$c_n(t, x) = (x - K)^+, \quad x \geq 0. \quad (21.49)$$

The function $c_n(t_n, x)$ is convex in x . This is well-known, but we establish it here anyway to demonstrate a method we need later. To show convexity in x , we show that, whenever $0 \leq x_1 \leq x_2$ and $0 \leq \lambda \leq 1$, we have

$$c_n(t_n, (1 - \lambda)x_1 + \lambda x_2) \leq (1 - \lambda)c_n(t_n, x_1) + \lambda c_n(t_n, x_2). \quad (21.50)$$

We begin with the observation that, for any number α , the function $(\alpha x - K)^+$ is convex in x , and therefore

$$\left(x \exp \left\{ \sigma(\widetilde{W}(T) - \widetilde{W}(t_n)) + \left(r - \frac{1}{2}\sigma^2 \right) (T - t_n) \right\} - K \right)^+$$

is convex in x . It follows that

$$\begin{aligned}
& c_n(t_n, (1 - \lambda)x_1 + \lambda x_2) \\
&= \widetilde{\mathbb{E}} \left[e^{-r(T-t_n)} \left(((1 - \lambda)x_1 + \lambda x_2) \exp \left\{ \sigma(\widetilde{W}(T) - \widetilde{W}(t_n)) \right. \right. \right. \\
&\quad \left. \left. \left. + \left(r - \frac{1}{2}\sigma^2 \right) \right\} - K \right)^+ \right] \\
&\leq (1 - \lambda) \widetilde{\mathbb{E}} \left[e^{-r(T-t_n)} \left(x_1 \exp \left\{ \sigma(\widetilde{W}(T) - \widetilde{W}(t_n)) + \left(r - \frac{1}{2}\sigma^2 \right) \right\} - K \right)^+ \right] \\
&\quad + \lambda \widetilde{\mathbb{E}} \left[e^{-r(T-t_n)} \left(x_2 \exp \left\{ \sigma(\widetilde{W}(T) - \widetilde{W}(t_n)) + \left(r - \frac{1}{2}\sigma^2 \right) \right\} - K \right)^+ \right] \\
&= (1 - \lambda)c_n(t_n, x_1) + \lambda c_n(t_n, x_2). \tag{21.51}
\end{aligned}$$

This proves [Equation \(21.50\)](#).

At time t_n , immediately before the dividend payment, the owner of the American call has two choices. She can exercise the option and receive $S(t_n-) - K$, or she can decline to exercise, permit the dividend to be paid (not to her) and the asset price to fall to $S(t_n) = (1 - a_n)S(t_n-)$, and have an option valued at $c_n(t_n, (1 - a_n)S(t_n-))$. The optimal decision is to exercise if $S(t_n-) - K > c_n(t_n, (1 - a_n)S(t_n-))$ and to decline to exercise if $S(t_n-) - K < c_n(t_n, (1 - a_n)S(t_n-))$. If $S(t_n-) - K = c_n(t_n, (1 - a_n)S(t_n-))$, it does not matter whether she exercises or declines to exercise. Therefore, the call value at time t_n immediately before the dividend is paid is $h_n(S(t_n-))$, where

$$h_n(x) = \max\{x - K, c_n(t_n, (1 - a_n)x)\}, \quad x \geq 0. \tag{21.52}$$

We show that $h_n(x)$ satisfies the assumptions of [Lemma 21.2](#). It is clear that $h_n(x) \geq 0$ for all $x \geq 0$ because $c_n(t_n, (1 - a_n)x) \geq 0$ for all $x \geq 0$. It is also clear that $h_n(0) = 0$ because $c_n(t_n, (1 - a_n)0) = 0$. To establish the convexity of $h_n(x)$, we recall from [Equation \(21.50\)](#) that $c_n(t_n, x)$ is convex in x . For $0 \leq x_1 \leq x_2$ and $0 \leq \lambda \leq 1$, we replace x_1 in [Equation \(21.50\)](#) by $(1 - a_n)x_1$ and replace x_2 by $(1 - a_n)x_2$ to obtain

$$c_n(t_n, (1 - a_n)((1 - \lambda)x_1 + \lambda x_2)) \leq (1 - \lambda)c_n(t_n, (1 - a_n)x_1) + \lambda c_n(t_n, (1 - a_n)x_2).$$

This shows that $c_n(t, (1 - a_n)x)$ is a convex function of x . The maximum of two convex functions is convex, and therefore $h_n(x)$ defined by Equation (21.52) is convex.

Starting from time t , where $t_{n-1} \leq t < t_n$, the owner of the American call can exercise at any time $u \in [t, t_n)$, and if she does, she receives $S(u) - K$. If she does not exercise prior to t_n , then at time t_n , immediately before the dividend payment, she owns a call whose value we have just determined to be $h_n(S(t_n-))$. Therefore, for $t_{n-1} \leq t < t_n$, the American call expiring at time T has the same price as the American call expiring immediately before the dividend payment at date t_n and paying $h_n(S(t_n-))$ upon expiration.

Because the underlying asset evolves as a geometric Brownian motion after the dividend is paid at time t_{n-1} until the dividend is paid at time t_n , Lemma 21.2 implies that $e^{-rt}h_n(S(t))$ is a submartingale for $t_{n-1} \leq t < t_n$.

In particular,

$$\tilde{\mathbb{E}} \left[e^{-r(u-t)} h_n(S(u)) \mid \mathcal{F}(t) \right] \geq h_n(S(t)), \quad t_{n-1} \leq t \leq u < t_n,$$

and letting $u \uparrow t_n$, we obtain

$$\tilde{\mathbb{E}} \left[e^{-r(t_n-t)} h_n(S(t_n-)) \mid \mathcal{F}(t) \right] \geq h_n(S(t)). \quad (21.53)$$

By the definition of $h_n(x)$,

$$h_n(S(t)) \geq S(t) - K. \quad (21.54)$$

This shows that the value of the European call expiring at time t_n immediately before the dividend is paid and paying $h_n(S(t_n-))$ upon expiration, which is the left-hand side of (21.53), is greater than or equal to the intrinsic value of the American call, which is the right-hand side of (21.54). Therefore, the option to exercise the American call before time t_n is worthless, and the American call value is the same as the value of the European call just described.

Because $S(t)$ is a Markov process, there is some function $c_{n-1}(t, x)$ such that the left-hand side of (21.53), the European call value, is

$$c_{n-1}(t, S(t)) = \tilde{\mathbb{E}} \left[e^{-r(t_n-t)} h_n(S(t_n-)) \mid \mathcal{F}(t) \right]. \quad (21.55)$$

The function $c_{n-1}(t, x)$ can be determined by computing the conditional expectation in (21.55) under the condition $S(t) = x$. In the case $t = t_{n-1}$, using Equation (21.41), this leads to

$$c_{n-1}(t_{n-1}, x) = \tilde{\mathbb{E}} \left[e^{-r(t_n-t_{n-1})} \times h_n \left(x \exp \left\{ \sigma(\tilde{W}(t_n) - \tilde{W}(t_{n-1})) + \left(r - \frac{1}{2}\sigma^2 \right) (t_n - t_{n-1}) \right\} \right) \right]. \quad (21.56)$$

The function $c_{n-1}(t, x)$ also satisfies the Black-Scholes-Merton differential equation

$$\frac{\partial}{\partial t} c_{n-1}(t, x) + rx \frac{\partial}{\partial x} c_{n-1}(t, x) + \frac{1}{2} \sigma^2 x^2 \frac{\partial^2}{\partial x^2} c_{n-1}(t, x) = r c_{n-1}(t, x), \quad t_{n-1} \leq t < t_n, \quad x \geq 0. \quad (21.57)$$

and the terminal condition

$$c_{n-1}(t_n, x) = h_n(t_n, x), \quad x \geq 0. \quad (21.58)$$

We repeat this process, defining

$$h_{n-1}(x) = \max\{x - K, c_{n-1}(t_{n-1}, (1 - a_{n-1})x)\}, \quad x \geq 0.$$

We can show as above that $h_{n-1}(x)$ satisfies the hypotheses of Lemma 8.5.1, and we continue.

In conclusion, we obtain an algorithm for the American call price on an asset paying dividends at the dates $t_1, t_2, \dots, t_n$. Solve recursively for $j = n, n-1, \dots, 0$, the partial differential equation

$$\frac{\partial}{\partial t} c_{j-1}(t, x) + rx \frac{\partial}{\partial x} c_{j-1}(t, x) + \frac{1}{2} \sigma^2 x^2 \frac{\partial^2}{\partial x^2} c_{j-1}(t, x) = r c_{j-1}(t, x), \quad t_{j-1} \leq t < t_j, \quad x \geq 0, \quad (21.59)$$

with the terminal condition

$$c_{j-1}(t_j, x) = h_j(x), \quad x \geq 0. \quad (21.60)$$

The functions $c_n(t, x)$ and $h_n(x)$ needed to get started are given by [Equations \(21.46\)](#) to [\(21.52\)](#), and the function $h_{j-1}(x)$ needed for the next step is given by

$$h_{j-1}(x) = \max\{x - K, c_{j-1}(t_{j-1}, (1 - a_{j-1})x)\}, \quad x \geq 0. \quad (21.61)$$

For $t_{j-1} \leq t < t_j$, if $S(t) = x$, then $c_{j-1}(t, x)$ is the American call price. Within each interval $[t_{j-1}, t_j)$, the American call price is actually the price of a European call expiring at t_j . The optimal exercise time is immediately prior to the dividend payment at the smallest time t_j for which $S(t_j-) - K$ exceeds $c_j(t_j, (1 - a_j)S(t_j-))$. If there is no t_j for which this condition is satisfied, then optimal exercise takes place at time T if $S(T) > K$, and otherwise the option should be allowed to expire unexercised.

21.4 Approximate Continuation Values

Regression-based methods posit an expression for the continuation value of the form

$$\mathbb{E}[V_{i+1}(X_{i+1}) \mid X_i = x] = \sum_{r=1}^M \beta_{ir} \psi_r(x), \quad (21.62)$$

for some basis functions $\psi_r : \mathbb{R}^d \rightarrow \mathbb{R}$ and constants β_{ir} , $r = 1, \dots, M$. Using the notation C_i for the continuation value at time i , we may equivalently write [Equation \(21.62\)](#) as

$$C_i(x) = \beta_i^\top \psi(x), \quad (21.63)$$

with

$$\beta_i^\top = (\beta_{i1}, \dots, \beta_{iM}), \quad \psi(x) = (\psi_1(x), \dots, \psi_M(x))^\top.$$

One could use different basis functions at different exercise dates, but to simplify notation, we suppress any dependence of ψ on i .

Assuming a relation of the form Equation (21.62) holds, the vector β_i is given by

$$\beta_i = (\mathbb{E}[\psi(X_i)\psi(X_i)^\top])^{-1}\mathbb{E}[\psi(X_i)V_{i+1}(X_{i+1})] \equiv B_\psi^{-1}B_{\psi V}. \quad (21.64)$$

Here, B_ψ is the indicated $M \times M$ matrix (assumed nonsingular) and $B_{\psi V}$ the indicated vector of length M . The variables (X_i, X_{i+1}) inside the expectations have a joint distribution corresponding to the state of the underlying Markov chain at dates i and $i + 1$.

The coefficients β_{ir} could be estimated from observations of pairs $(X_{ij}, V_{i+1}(X_{i+1,j}))$, $j = 1, \dots, b$, each consisting of the state at time i and the corresponding option value at time $i + 1$. Consider, in particular, independent paths $(X_{1j}, \dots, X_{mj})$, $j = 1, \dots, b$, and suppose for a moment that the values $V_{i+1}(X_{i+1,j})$ are known. The least-squares estimate of β_i is then given by

$$\hat{\beta}_i = \hat{B}_\psi^{-1}\hat{B}_{\psi V},$$

where $\hat{B}_\psi$ and $\hat{B}_{\psi V}$ are the sample counterparts of B_ψ and $B_{\psi V}$. More explicitly, $\hat{B}_\psi$ is the $M \times M$ matrix with qr entry

$$\frac{1}{b} \sum_{j=1}^b \psi_q(X_{ij})\psi_r(X_{ij})$$

and $\hat{B}_{\psi V}$ is the M -vector with r th entry

$$\frac{1}{b} \sum_{k=1}^b \psi_r(X_{ik})V_{i+1}(X_{i+1,k}). \quad (21.65)$$

All of these quantities can be calculated from function values at pairs of consecutive nodes $(X_{ij}, X_{i+1,j})$, $j = 1, \dots, b$. In practice, V_{i+1} is unknown and must be replaced by estimated values $\hat{V}_{i+1}$ at downstream nodes. The estimate $\hat{\beta}_i$ then defines an estimate

$$\hat{C}_i(x) = \hat{\beta}_i^\top \psi(x), \quad (21.66)$$

of the continuation value at an arbitrary point x in the state space $\mathbb{R}^d$. This, in turn, defines a procedure for estimating the option value.

Algorithm 1 Regression-Based Pricing Algorithm

- 1: Simulate b independent paths $\{X_{1j}, \dots, X_{mj}\}$, $j = 1, \dots, b$, of the Markov chain.
- 2: At terminal nodes, set $\hat{V}_{mj} = h_m(X_{mj})$ for $j = 1, \dots, b$.
- 3: **for** $i = m - 1$ **down to** 1 **do**
- 4: Given estimated values $\hat{V}_{i+1,j}$, $j = 1, \dots, b$, use regression to calculate $\hat{\beta}_i = \hat{B}_\psi^{-1} \hat{B}_\psi V$.
- 5: Compute estimated continuation values $\hat{C}_i(X_{ij})$ using Equation (21.66).
- 6: **for** $j = 1, \dots, b$ **do**
- 7: Set value based on immediate exercise versus estimated continuation:

$$\hat{V}_{ij} = \max \{h_i(X_{ij}), \hat{C}_i(X_{ij})\}. \quad (21.67)$$

- 8: **end for**
- 9: **end for**
- 10: Set $\hat{V}_0 = (\hat{V}_{11} + \dots + \hat{V}_{1b})/b$.

Definition 21.2: American Contingent Claim

An *American contingent claim* $X^a = (X, \mathcal{T}_{[0,T]})$ expiring at time T consists of a sequence of payoffs $(X_t)_{t=0}^T$, where X_t is an $\mathcal{F}_t$ -measurable random variable for $t = 0, 1, \dots, T$, and of the set $\mathcal{T}_{[0,T]}$ of admissible exercise policies.

Proposition 21.1: Arbitrage Price of American Put

The arbitrage price P_t^a of an American put option equals

$$P_t^a = \max_{\tau \in \mathcal{T}_{[t,T]}} \mathbb{E}_{\mathbb{P}^*} \left(\hat{r}^{-(\tau-t)} (K - S_\tau)^+ \mid \mathcal{F}_t \right), \quad \forall t \leq T. \quad (21.68)$$

Moreover, for any $t \leq T$ the stopping time τ_t^* that realizes the maximum in Equation (21.68) is given by the expression (by convention $\min \emptyset = T$)

$$\tau_t^* = \min \{u \in \{t, t+1, \dots, T\} \mid (K - S_u)^+ \geq P_u^a\}. \quad (21.69)$$

Proposition 21.2: Arbitrage Price of American Claim in CRR

For every $t \leq T$, the arbitrage price $\pi(X^a)$ of an arbitrary American claim X^a in the CRR model equals

$$\pi_t(X^a) = \max_{\tau \in \mathcal{T}_{[t,T]}} \mathbb{E}_{\mathbb{P}^*} \left(\hat{r}^{-(\tau-t)} X_\tau \mid \mathcal{F}_t \right). \quad (21.70)$$

The price process $\pi(X^a)$ can be determined using the following recurrence relation, for every $t \leq T - 1$,

$$\pi_t(X^a) = \max \left\{ X_t, \mathbb{E}_{\mathbb{P}^*} \left(\hat{r}^{-1} \pi_{t+1}(X^a) \mid \mathcal{F}_t \right) \right\} \quad (21.71)$$

subject to the terminal condition $\pi_T(X^a) = X_T$. In the case of a path-independent American claim X^a with the reward function h we have that, for every $t \leq T - 1$,

$$\pi_t(X^a) = \max \left\{ h(S_t, t), \hat{r}^{-1} \left(p_* \pi_{t+1}^u(X^a) + (1 - p_*) \pi_{t+1}^d(X^a) \right) \right\}, \quad (21.72)$$

where, for a generic stock price value S_t at time t , we write $\pi_{t+1}^u(X^a)$ and $\pi_{t+1}^d(X^a)$ to denote the values of the price process $\pi(X^a)$ at time $t + 1$ in the nodes that correspond to the upward and downward movements of the stock price during the time-period $[t, t + 1]$, that is, for the values uS_t and dS_t of the stock price at time $t + 1$, respectively.

22 Exotic Options

23 Basket Options

Appendices

A Existence of Lebesgue Measure

B Linear Algebra and Functional Analysis

C Multivariate Differentiation

D Probability and Statistics

E Python Primer